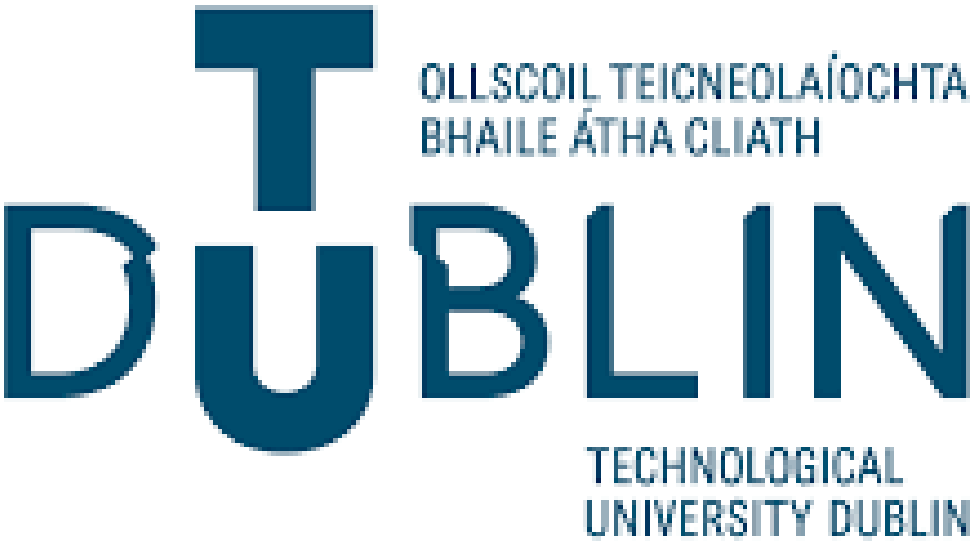


1.0 Overview

With an industrial kitchen / restaurant, it is important that the correct systems are in place to allow the business to be profitable. Conversely, this is not a cheap endeavour. There needs to be a balance struck between cost, overheads and ultimately potential profit.

This is where a suggestion of a unique IoT focused kitchen installation is a perfect solution. Allowing for kitchen staff to be more efficient with the management of kitchen and restaurant materials.

Ultimately, the aim was to make the business profitable by balancing costs with profits across the various roles within the business.

2.0 Objectives or Scope

For the project with 2 areas of responsibility each, it amounted to 6 departments. These were:

Project Manufacturing:

- o Four sub focus areas.
- o Component Manufacturing.
  - § This covers logistical planning, cost management, and supplier evaluation.
- o Quality Assurance.
  - § This covers vital areas such as system integrity, performance under load, and final QA testing.
- o IoT Hardware Manufacturing.
  - § This covers the development and manufacture of the sensor components, and the successful integration into the final product.
- o The Production aspect of manufacturing.
  - § This ensures that the production is logical, modular, staff are correctly trained to be safe, efficient and reliable.

Project Support:

- o Four sub focus areas.
- o Pre-Install support
  - § Covers site assessment, System customisation, and preparation of the space
- o User Training
  - § Comprises of hands-on training, Online resource creation, and ensuring there is support available if needed.
- o Post install support
  - § Covers remote monitoring of systems, troubleshooting and resolving any detected faults / issues.

§ Maintenance in place covering software updates, as well as preventative / scheduled maintenance.

Technical Installation:

Responsible for the installation of the equipment on site.

- Equipment Placement

  Clearing the installation site, measuring the dimensions, transporting and arranging equipment on site.

- Plumbing Install

  Connecting the equipment to site plumbing.

- Electrical Install

  Connecting the equipment to site mains electricity.

  Ensure compliance with regulations.

- Network Connectivity

  Configuring the IoT-enabled equipment to connect to the site network.

  Setup remote monitoring capabilities.

- On-site testing

  Testing the equipment to ensure functionality and compliance.

  Provide client with training on proper equipment use.

Marketing:

- Market Research
  Conducting surveys and identifying consumer demands.

- Campaign Design

  Creating promotional materials including brochures and digital marketing.

Running marketing campaigns across various platforms.

- Trade Show

  Attending trade shows, planning booth design, logistics, and promotional activities.

- Customer outreach

  Tracking marketing effectiveness and getting customer feedback to refine marketing strategies.

HR:

- Ensure successful recruitment, training, and employee wellbeing.
- Define roles and responsibilities, ensuring the right skill sets are in place
- Manage the recruitment process to hire qualified individuals.
- Develop training programs that align with organizational goals.

R&D:

- Drive innovation and continuous improvement in products and systems
- Design and develop IOT-enabled solutions that meet market needs and technological standards.
- Research emerging trends and incorporate them into product development

3.0 General approach

The project's General Approach is centred around effective collaboration between the R&D and HR teams to ensure that the system is not only technologically advanced but also supported by a skilled workforce. The approach can be broken down as follows:

1. Phase 1: Research & Development (R&D)

- Design and Innovation: The R&D team will be tasked with designing the IoT components of the kitchen system, ensuring that they are compatible with existing kitchen infrastructure and scalable to different commercial premises.
- Prototyping and Testing: The team will create and test functional prototypes of the IoT-enabled components, refining designs based on feedback from internal stakeholders, engineers, and users.
- System Integration: The R&D department will also work closely with the technical installation team to ensure that all IoT devices are seamlessly integrated into the commercial kitchen systems. This will include extensive on-site testing and troubleshooting.

2. Phase 2: Human Resources (HR)

- Recruitment: HR will focus on recruiting skilled technicians, R&D personnel, and support staff who have the expertise in IoT, commercial kitchen technology, and system integration. Detailed job descriptions, recruitment strategies, and interview processes will be defined to ensure that the right talent is brought on board.

- Training and Development: Once employees are hired, HR will coordinate comprehensive training sessions on IoT technology, safety protocols, and kitchen systems to enhance team competency and ensure high-quality project execution.
- Conflict Resolution and Safety Management: HR will also ensure that proper conflict resolution procedures are in place to maintain a positive and productive work environment. Additionally, HR will work with the safety manager to establish safety standards that align with industry regulations to protect staff and ensure safe installations.

3. Phase 3: Ongoing Support and Maintenance (Post-Installation)

- While not directly in charge of technical support, the HR team will ensure that the appropriate personnel are assigned to maintenance and support roles. Ongoing training will be necessary to ensure the workforce is prepared to address issues related to IoT systems and offer effective troubleshooting.
- R&D will continue to support the project after the installation phase, assisting with software upgrades, system diagnostics, and improvements based on user feedback.

By combining these two areas—Research & Development and Human Resources—the project ensures both the innovation of the product and the smooth execution by a highly skilled team. Effective collaboration between the two teams is essential for the timely development and installation of the IoT-enabled commercial kitchen system.

3.1 Stakeholder Identification

Stakeholder identification is a critical aspect of managing the IoT-enabled Commercial Kitchen System project, ensuring that all parties with a personal stake in the project are properly acknowledged and engaged. The stakeholders can be categorized into internal and external groups, with each group playing a key role in the success of the project.

Internal Stakeholders:

1. R&D Team:

- Responsible for the design, testing, and integration of IoT components into the kitchen system.
- Directly impacts the functionality and innovation of the project.

2. HR Team:

- Responsible for recruiting, training, and managing personnel.
- Ensures that the project is staffed with skilled technicians, project managers, and support staff.

3. Project Manager:

- Oversees the overall project execution, ensuring timelines and deliverables are met.
- Serves as a liaison between all departments.

4. Installation and Technical Teams:

- In charge of the physical installation of the IoT-enabled kitchen systems and on-site troubleshooting.
- Close coordination with the R&D department for system integration.

5. Marketing Team:

- Responsible for promoting the IoT system, managing industry expos, and conducting market research.

6. Safety and Compliance Team:

- Ensures that safety regulations are met during installation and operation.
- Collaborates with HR to ensure employee safety standards are in place.

External Stakeholders:

1. Suppliers and Manufacturers:

- Provide the components and equipment necessary for the development and installation of the IoT systems.
- R&D will work closely with suppliers to ensure component quality.
- Customers (Commercial Kitchen Operators):
- End-users of the IoT-enabled kitchen system, including businesses in hospitality, canteens, and large-scale commercial kitchens.
- Feedback from customers is crucial for product refinement and ongoing maintenance.

2. Regulatory Bodies:

- Ensure that the IoT system complies with industry-specific regulations, such as safety and energy efficiency standards.
- Work closely with the safety and compliance team for audits and certifications.

3. External Consultants:

- Legal advisors and technical consultants who may offer expertise during the design, installation, and safety phases.
- Ensuring compliance with local laws and regulations related to the IoT system.

3.2 Stake Holder Communication

Stakeholders are a key part of the project. They will be kept abreast of any developments and updated monthly. This is to ensure that they are happy with the direction in which the project is progressing. They will be given multiple methods of communication in addition to the monthly updates. Any key decisions for the direction of the business will involve the stakeholders being contacted and consulted with.

4.0 Contractual aspects

The contractual elements outline responsibilities, expectations, and protections for all stakeholders involved in the IoT-enabled Commercial Kitchen System project.

4.1 Reporting Requirements

- Internal: Monthly progress reports detailing milestones, issues, and resource allocation.
- External: Monthly updates to customers and quarterly updates to suppliers and regulatory bodies, covering project progress, delivery, and testing.
- Performance Reviews: Quarterly internal and external reviews to assess progress and address concerns.

4.2 Customer-Supplied Resources

- Equipment & Infrastructure: Customers must ensure the provision of adequate space, electrical connections, and plumbing for system installation.
- Personnel: If applicable, customers should provide trained staff for the installation process.

4.3 Liaison Arrangement

- Project Liaison: The Project Manager serves as the main point of contact for all stakeholders.
- Escalation: An escalation process will be followed for unresolved issues, involving senior management when necessary.

4.4 Advisory Committees

- Project Advisory Committee: Includes key internal stakeholders (e.g., Project Manager, R&D Lead) to review progress and offer strategic advice.
- Technical & Safety Committees: Focus on technical support and safety compliance, respectively.

4.5 Project Review and Cancellation

- Regular Reviews: Quarterly reviews with stakeholders to assess progress.
- Cancellation Clause: Provisions for project cancellation with a 30–60-day notice period and compensation for completed work.

4.6 Proprietary Requirements

- IP Ownership: The company retains ownership of developed IoT technology, with customers receiving usage rights through licensing agreements.
- Confidentiality: Contracts include clauses to protect proprietary information.

4.7 Specific Management Agreements

- Project Management: Defines the Project Manager's responsibilities for overseeing the project, ensuring timelines are met, and coordinating resources.

- R&D Management: Specifies goals for developing and integrating the IoT technology.
- HR Management: Clarifies HR roles in recruitment, training, and ensuring safety and compliance.

5.0 Project Planning

5.1 Mind map

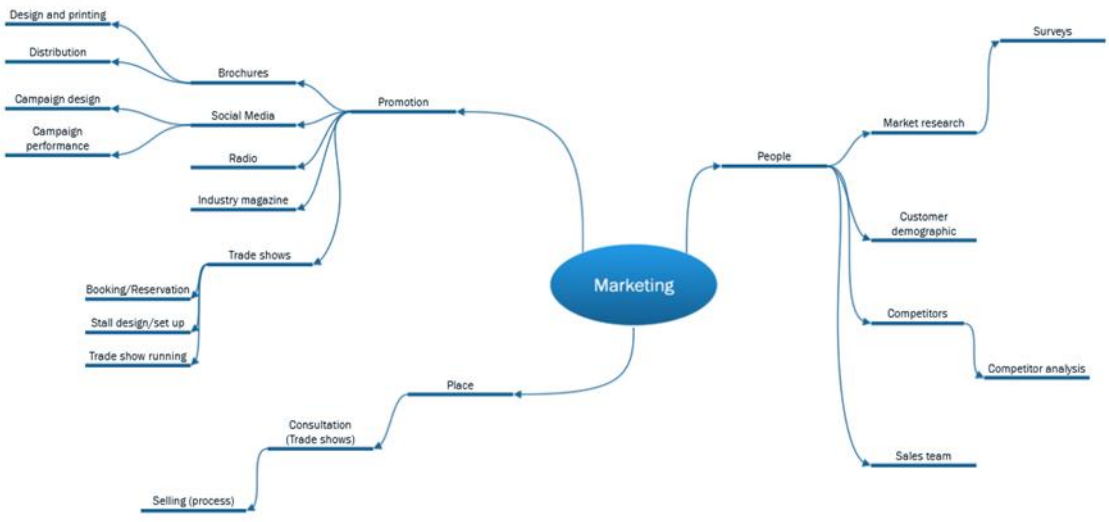


Figure 1 Marketing mind map

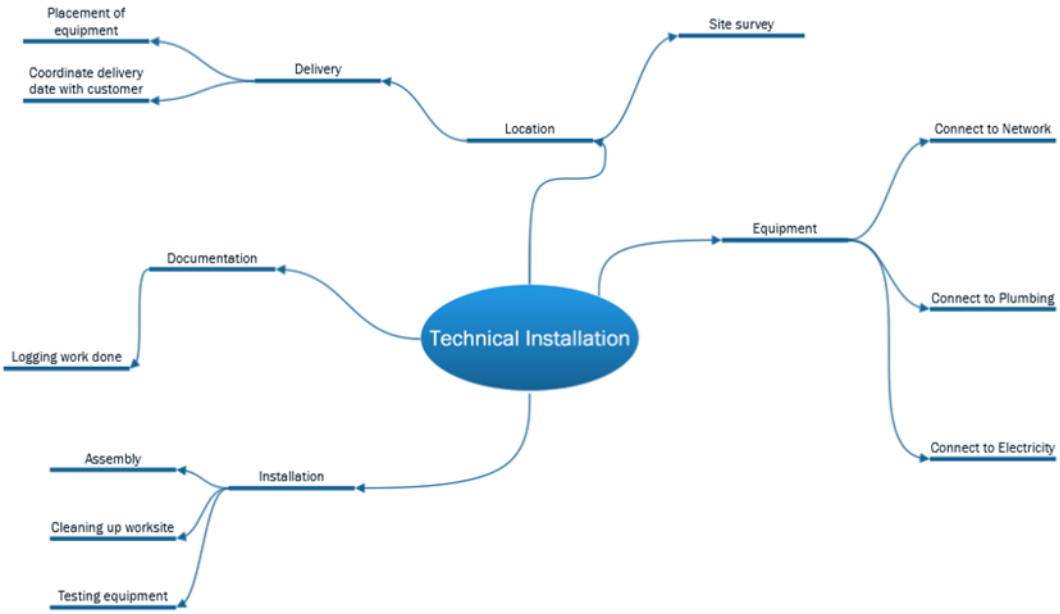


Figure 2 Technical Installation mind map

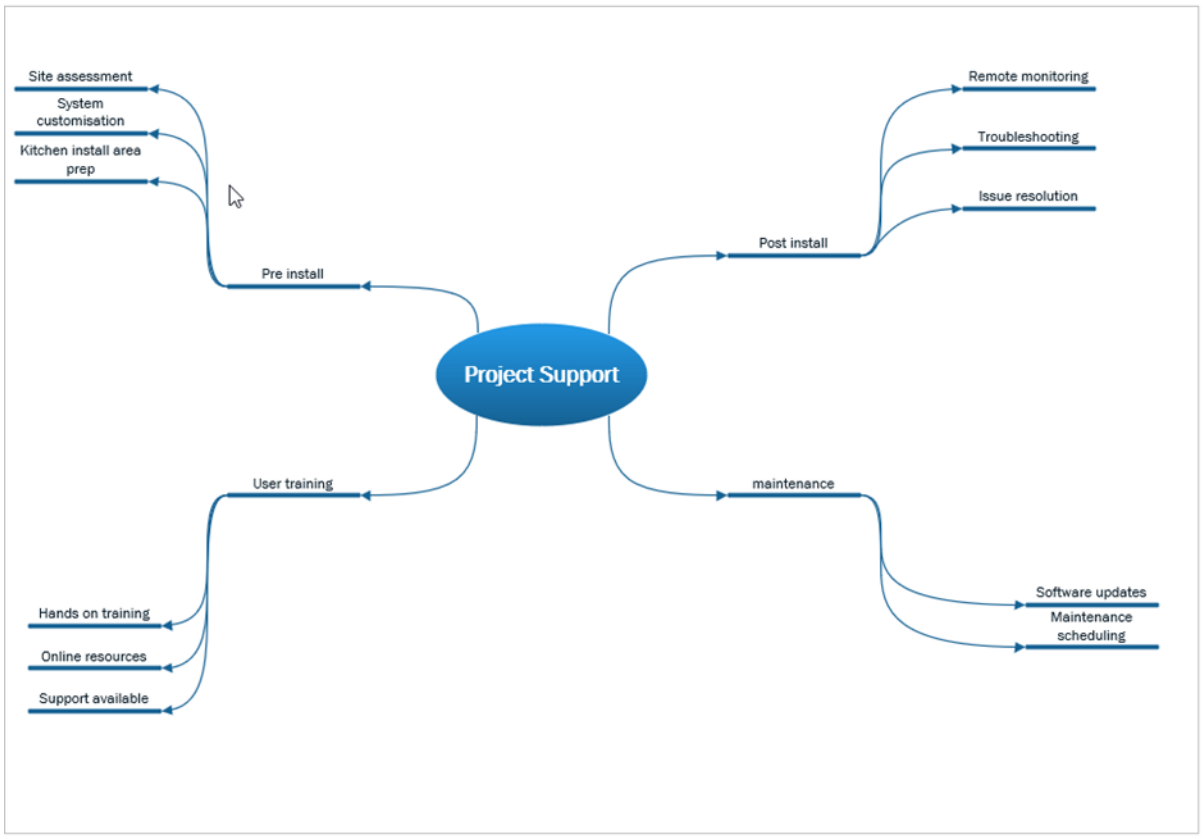


Figure 3 Project Support mind map

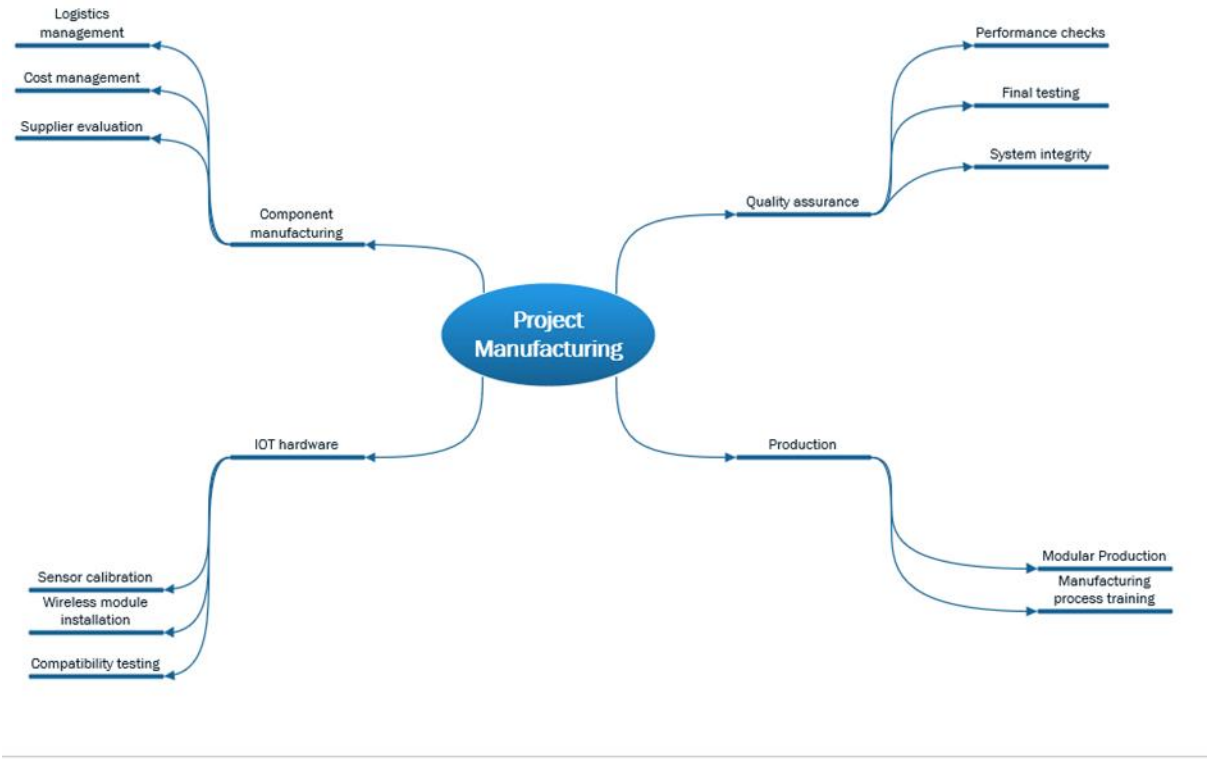


Figure 4 Project Manufacturing mind map

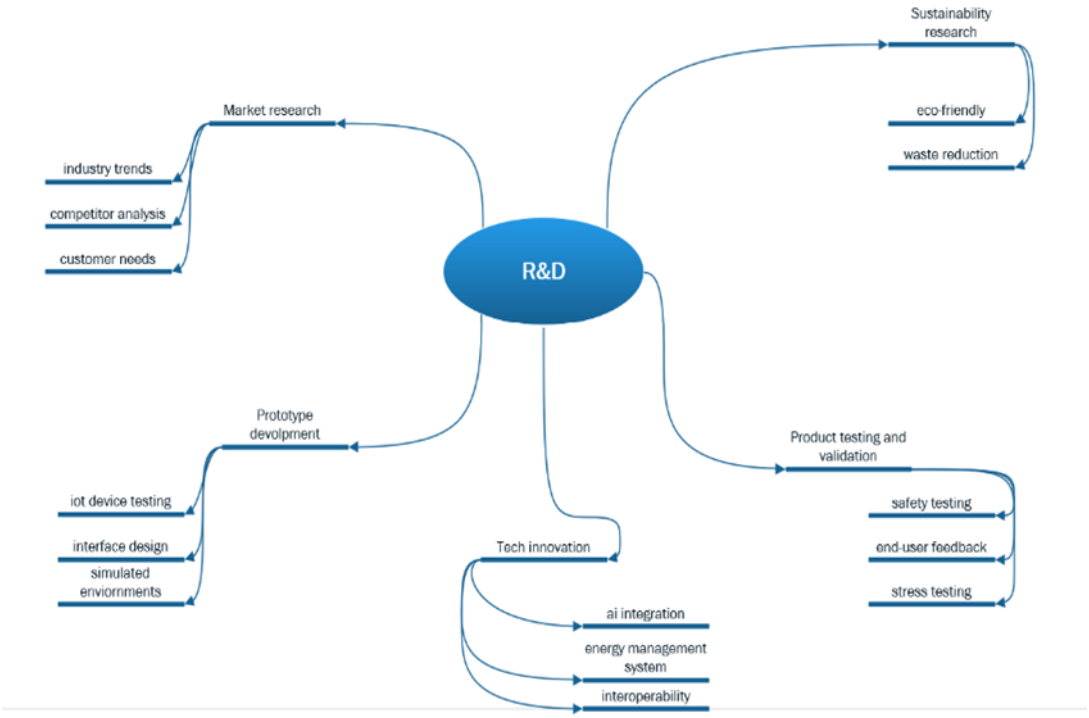


Figure 5 R&D mind map

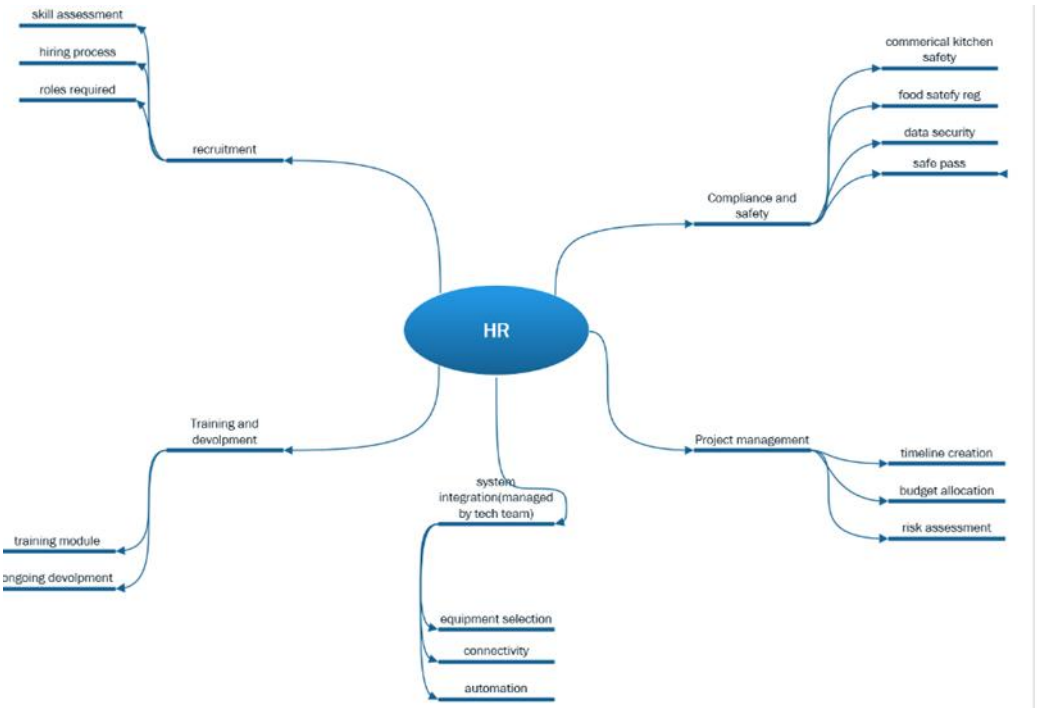


Figure 6 HR mind map

5.2 Work Breakdown Structures

5.2 recruitment

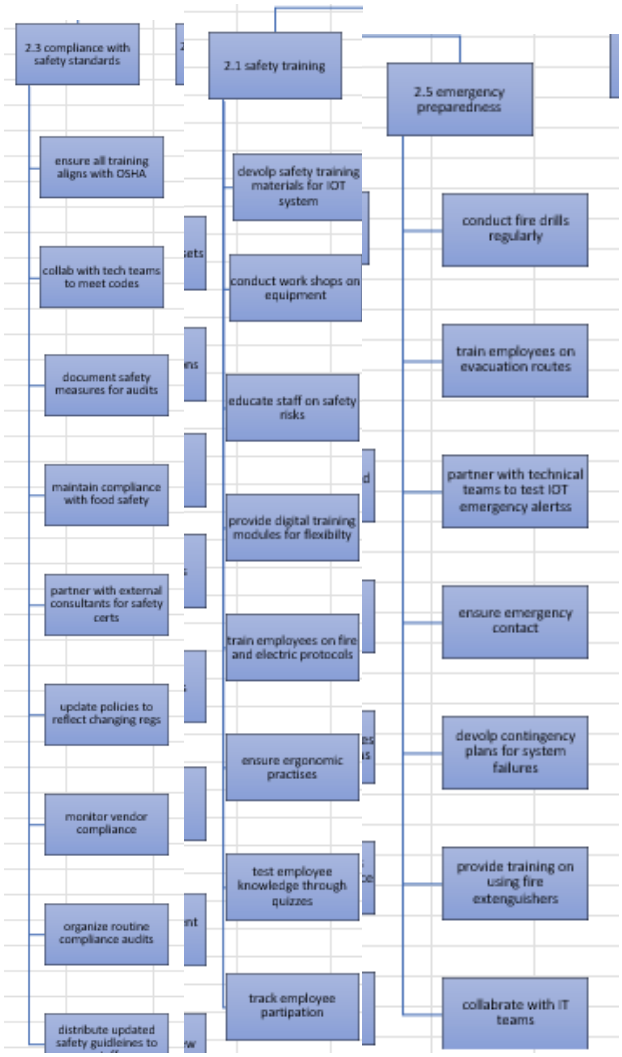


FIGURE 3

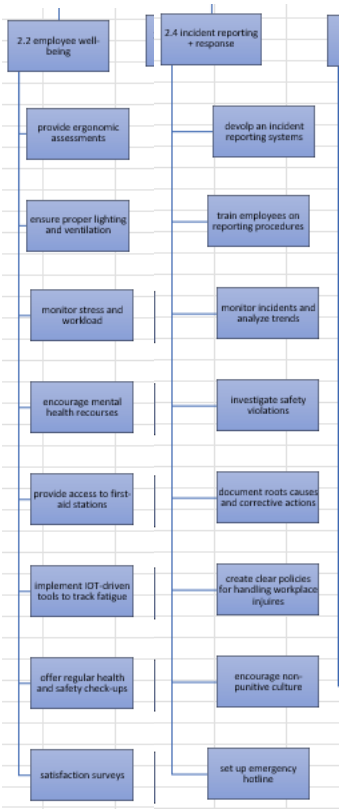
5.3 safety


2.0 safety
2.1 safety training
2.2 employee well-being
2.3 compliance with safety standards
2.4 incident reporting + response
2.5 emergency preparedness
devolp safety training materials for IOT system
conduct work shops on equipment
educate staff on safety risks
provide digital training modules for flexibility
train employees on fire and electric protocols
ensure ergonomic practises
test employee knowledge through quizzes
track employee partipation
provide ergonomic assessments
ensure proper lighting and ventilation
monitor stress and workload
encourage mental health recourses
provide access to first-aid stations
implement IOT-driven tools to track fatigue
offer regular health and safety check-ups
satisfaction surveys
ensure all training aligns with OSHA
collab with tech teams to meet codes
document safety measures for audits
maintain compliance with food safety
partner with external consultants for safety certs
update policies to reflect changing regs
monitor vendor compliance
organize routine compliance audits
distribute updated safety guidelines to staff
devolp an incident reporting systems
train employees on reporting procedures
monitor incidents and analyze trends
investigate safety violations
document roots causes and corrective actions
create clear policies for handling workplace injures
encourage non-punitive culture
set up emergency hotline
conduct fire drills regularly
train employees on evacuation routes
partner with technical teams to test IOT emergency alertss
ensure emergency contact
devolp contingency plans for system failures
provide training on using fire extenguishers
collabrate with IT teams

FIGURE 4



(these three will be combined together)



(and these two will be combined)

FIGURE 5

5.4 training and development



(these two will be combined)

FIGURE 6

5.5 system integration



FIGURE 7

(these two will be combined)

5.6 project management

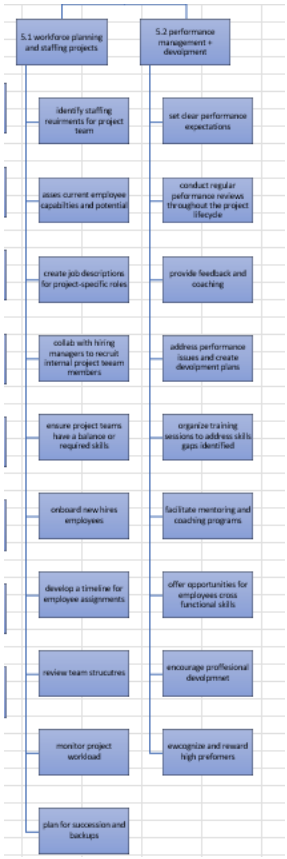


FIGURE 8

(these two will be combined)

R&D:

5.7 market research



FIGURE 10

5.8 technology innovation



FIGURE 11

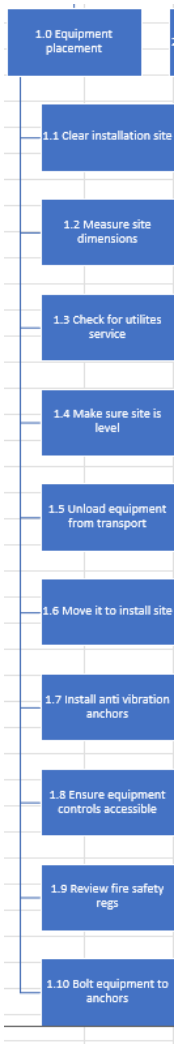
5.9 prototyping and product testing



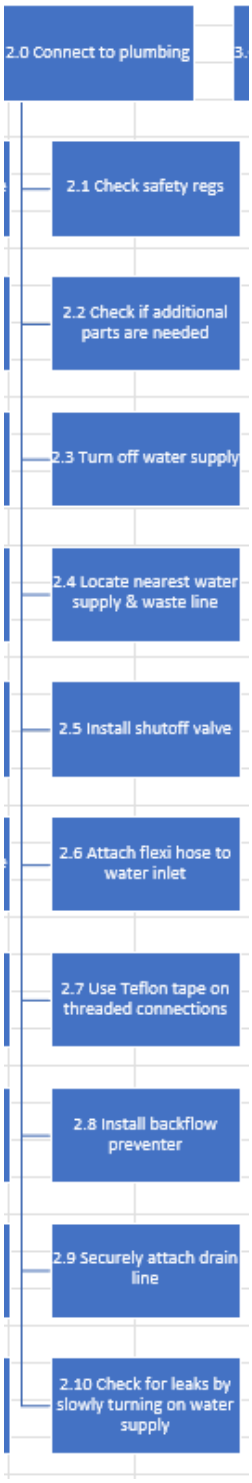
FIGURE 12

Technical Installation:

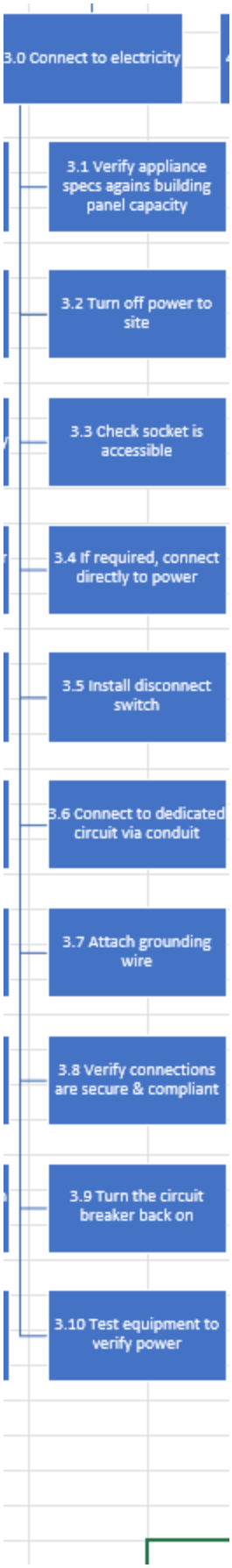
5.10 Equipment placement



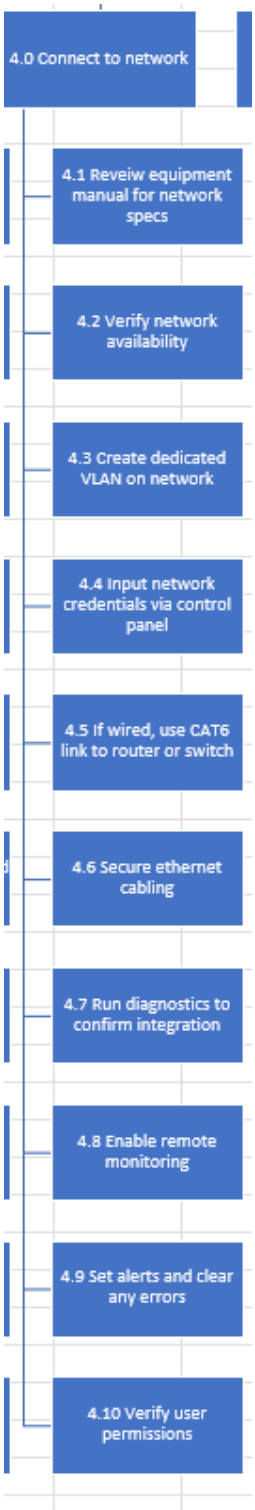
5.11 Connect to Plumbing



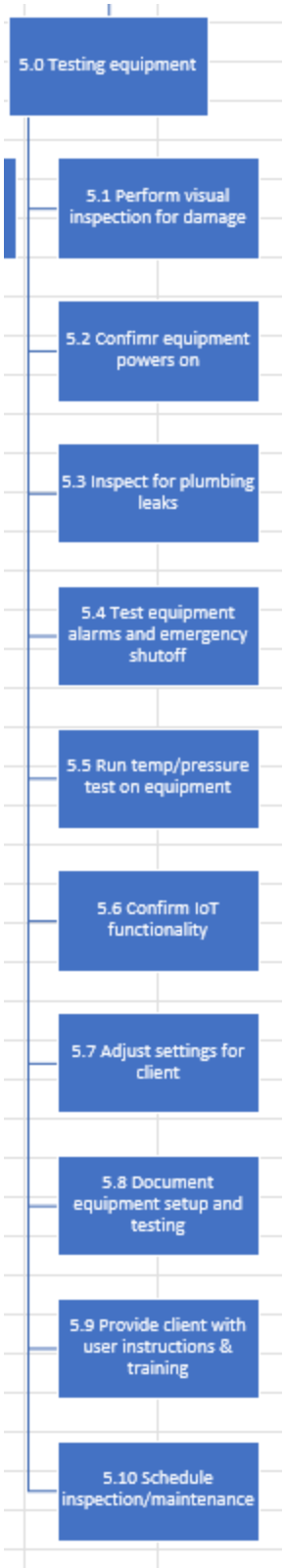
5.12 Connect to electricity


3.0 Connect to electricity
3.1 Verify appliance specs agains building panel capacity
3.2 Turn off power to site
3.3 Check socket is accessible
3.4 If required, connect directly to power
3.5 Install disconnect switch
3.6 Connect to dedicated circuit via conduit
3.7 Attach grounding wire
3.8 Verify connections are secure & compliant
3.9 Turn the circuit breaker back on
3.10 Test equipment to verify power

5.13 Connect to network

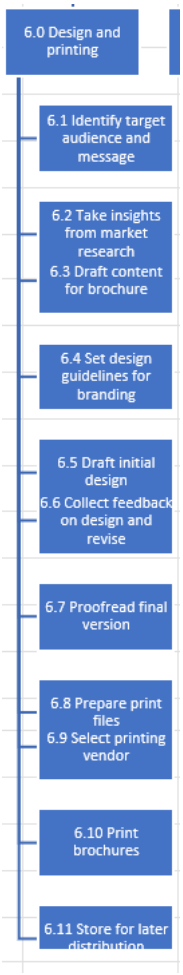


5.14 Testing equipment

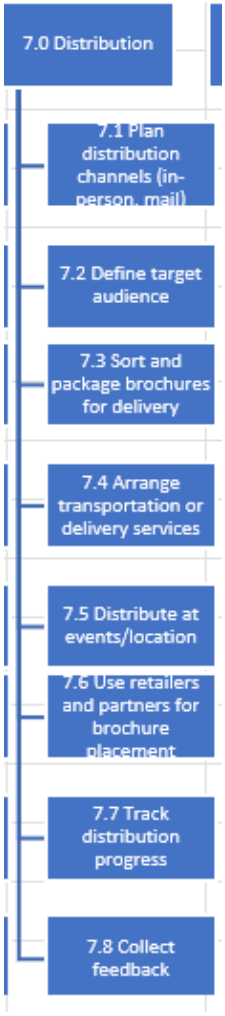


Marketing:

5.15 Design and printing



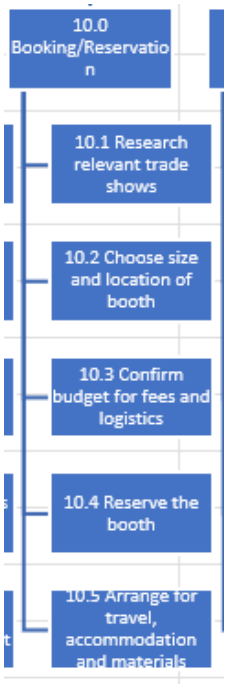
5.16 Distribution



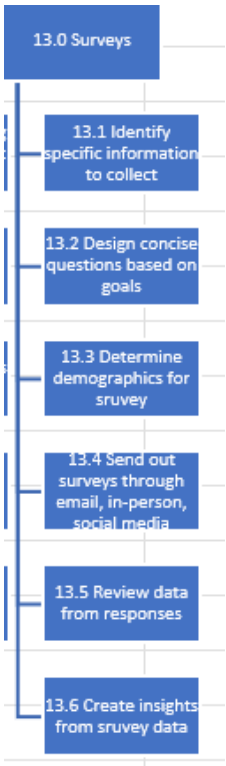
5.17 Campaign design



5.18 Booking/Reservation


10.0 Booking/Reservation
10.1 Research relevant trade shows
10.2 Choose size and location of booth
10.3 Confirm budget for fees and logistics
10.4 Reserve the booth
10.5 Arrange for travel, accommodation and materials

5.19 Surveys


13.0 Surveys
13.1 Identify specific information to collect
13.2 Design concise questions based on goals
13.3 Determine demographics for sruvey
13.4 Send out surveys through email, in-person, social media
13.5 Review data from responses
13.6 Create insights from sruvey data

5.3 Action Plans

Action Plan for Work Package 1.0: Site Assessment					
Deliverables on completion of Work Package: A plan for upgrade will be completed					
Measures of Accomplishments / Milestones Site is surveyed, and utilities are checked.					
Key constraints and Assumptions There is already a non IOT solution in place					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
1.0	Site assessment				
1.1	Understand Requirements			3 days	IOT solutions architect, Project manager
1.2	Evaluate existing space		1.1	2 days	Site engineer
1.3	Check the layout		1.2	1 day	Site engineer
1.4	Evaluate environment		1.3	2 days	Design engineer
1.5	Check utilities	Tool kit	1.4	1 day	Electrician, Plumber
1.6	Test Connectivity	Laptop	1.5	2 days	IT technician
1.7	Compliance Review		1.6	2 days	Compliance officer
1.8	Integration with existing systems		1.7	3 days	Systems engineer
1.9	Document plan		1.8	2 days	Project Manager

Action Plan for Work Package 2.0: System Customisation					
Deliverables on completion of Work Package: System will be planned to the need of customer.					
Measures of Accomplishments / Milestones Project planning after visit has taken place.					
Key constraints and Assumptions Every kitchen will have different requirements and layout.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
2.0	System Customisation				
2.1	Analyse flow of kitchen space			2 days	Site engineer

2.2	Ask about IOT requirements		2.1	1 day	IOT specialist
2.3	Configure Wash cycles		2.2	2 days	Software engineer
2.4	Plan for energy usage		2.3	2 days	Energy engineer
2.5	Future planning		2.4	3 days	Project manager
2.6	Document chosen settings		2.5	1 day	Kitchen manager

Action Plan for Work Package 3.0: Kitchen install prep					
Deliverables on completion of Work Package: The area will be ready for the installation to take place					
Measures of Accomplishments / Milestones Utilities have been checked. Installation area is clear					
Key constraints and Assumptions Proper precautions have been taken without disrupting existing plans.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
3.0	Kitchen install prep				
3.1	Plan equipment positions			2 days	Design engineer
3.2	Consult structural plans		3.1	1 day	Structural engineer
3.3	Plan for ventilation		3.2	1 day	HVAC engineer
3.4	Check water supply		3.3	1 day	Plumber
3.5	Plan electricity runs		3.4	2 days	Electrician
3.6	Clean installation areas		3.5	1 day	Kitchen cleaner
3.7	Clear installation areas		3.6	1 day	Maintenance engineer

Action Plan for Work Package 4.0: Online Resources					
Deliverables on completion of Work Package: Online training resources are available.					
Measures of Accomplishments / Milestones Training structure in place. Online platform is available.					
Key constraints and Assumptions The material must be mobile friendly to facilitate learning at work.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
4.0	Online Resources				

4.1	Identify training areas			2 days	Training Manager
4.2	Develop training structure		4.1	4 days	Training manager
4.3	Create video content		4.2	7 days	IOT dishwasher expert
4.4	Develop user manual		4.3	4 days	Product specialist
4.5	Provide troubleshooting steps		4.4	2 days	Support engineer
4.6	Use online platform		4.5	3 days	Learning management system admin
4.7	Mobile friendly content		4.6	2 days	UX designer

Action Plan for Work Package 5.0: Hands on Training					
Deliverables on completion of Work Package: Staff have been trained in machine systems					
Measures of Accomplishments / Milestones Common issues have been discussed					
Key constraints and Assumptions All kitchen and management staff must learn how systems operate.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
5.0	Hands on training				
5.1	Set up training room			1 day	Training coordinator
5.2	System overview		5.1	½ a day	Product specialist
5.3	Basic operation of machine		5.2	½ a day	Product specialist
5.4	Introduce IOT features		5.3	½ a day	IOT specialist
5.5	Common issues		5.4	1 day	Support engineer
5.6	Maintenance Training		5.5	2 days	Support engineer
5.7	Answer questions		5.6	½ a day	Product specialist
5.8	Post training support		5.7	ongoing	Support engineer

Action Plan for Work Package 6.0: Software Updates
Deliverables on completion of Work Package: Software updates for the system will have been performed
Measures of Accomplishments / Milestones Current configuration has been backed up.
Key constraints and Assumptions The machine must always have internet connectivity.

Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
6.0	Software Updates				
6.1	Schedule regular updates			1 day	IT admin
6.2	Review notes		6.1	1 day	IOT specialist
6.3	Back-up system		6.2	½ a day	Support engineer
6.4	Check update Requirements		6.3	½ a day	Software engineer
6.5	Test the update		6.4	2 days	IOT specialist
6.6	Apply the update		6.5	1 day	Software engineer
6.7	Validate update		6.6	½ a day	IOT specialist
6.8	Monitor system status		6.7	2 days	Support engineer
6.9	Document updates performed		6.8	½ a day	IT admin

Action Plan for Work Package 7.0: Maintenance Scheduling					
Deliverables on completion of Work Package: Routine maintenance will be scheduled.					
Measures of Accomplishments / Milestones Any faults have been identified and repaired.					
Key constraints and Assumptions All incidents and fixes must be documented.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
7.0	Maintenance Scheduling				
7.1	Detect issues			1 day	IOT Specialist
7.2	Issue reporting system		7.1	1 day	IT admin
7.3	Inspect the system		7.2	1 day	Maintenance technician
7.4	Perform repairs		7.3	2 days	Maintenance technician
7.5	Test the system		7.4	1 day	Support engineer
7.6	Document incidents		7.5	1 day	Maintenance supervisor
7.7	Schedule routine checks		7.6	½ a day	Support engineer

Action Plan for Work Package 8.0: Logistics Management
Deliverables on completion of Work Package: There must be transportation of assets available.
Measures of Accomplishments / Milestones Warehouse management system is online. Quality assurance is in place.
Key constraints and Assumptions

There will be contracted drivers who can also deliver equipment directly to site.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
8.0	Logistics Management				
8.1	Supply Chain Assessment			3 days	Logistics manager
8.2	Vendor Coordination		8.1	2 days	Procurement specialist
8.3	Means of Transport		8.2	1 day	Transport coordinator
8.4	Warehouse Management		8.3	2 days	Warehouse manager
8.5	Material Training		8.4	4 days	Warehouse supervisor
8.6	Production line Management		8.5	2 days	Logistics manager
8.7	Quality assurance		8.6	3 days	Warehouse manager
8.8	Distribution / delivery		8.7	2 days	Contracted drivers

Action Plan for Work Package 9.0:					
Deliverables on completion of Work Package: Hardware will have been designed and manufactured					
Measures of Accomplishments / Milestones IOT components created.					
Key constraints and Assumptions Each solution will be manufactured to the needs of the customer.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
9.0	Hardware Manufacture				
9.1	Solution design			4 days	Design engineer
9.2	Component Sourcing		9.1	6 days	Supply chain manager
9.3	Equipment Setup		9.2	3 days	Production manager
9.4	Material Preparation		9.3	2 days	Production manager
9.5	IOT circuit design		9.4	4 days	Electrical Engineer
9.6	IOT sensor integration		9.5	3 days	Electrical Engineer
9.7	Manufacturing of machine		9.6	8 days	Assembly worker
9.8	Hardware testing		9.7	4 days	Hardware Engineer
9.9	Hardware compliance checks		9.8	3 days	Compliance officer

Action Plan for Work Package 10.0: Cost Management					
Deliverables on completion of Work Package: Budget will have been completed.					
Measures of Accomplishments / Milestones Labour cost will have been established. Costs will be tracked.					
Key constraints and Assumptions Risk must be budgeted for. If there is a component shortage / dip in the hospitality industry.					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
10.0	Cost Management				
10.1	Budget Allocation			3 days	Accountant
10.2	Cost estimation per area		10.1	2 days	Project manager
10.3	Vendor Partnerships		10.2	4 days	Procurement manager
10.4	Cost of labour		10.3	2 days	Accountant
10.5	Energy and Utilities		10.4	2 days	Operations manager
10.6	Environmental initiatives		10.5	2 days	Operations manager
10.7	Cost tracking system		10.6	4 days	IT admin
10.8	Lifecycle cost		10.7	3 days	Project Manager
10.9	Risk mitigation budgeting		10.8	4 days	Risk manager

Action Plan for Work Package 1.0: Equipment Placement					
Deliverables on completion of Work Package: Site has been cleared and equipment placed					
Measures of Accomplishments / Milestones Site has been surveyed and prepped Equipment has been delivered and placed on site					
Key constraints and Assumptions Assume customer has made order and scheduled delivery					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
1.0	Equipment placement				
1.1	Clear installation site			2 hours	Technical delivery operator 1
1.2	Measure site dimensions	IT Administrator	1.1	1 hour	Technical delivery operator 1
1.3	Check for utilities service	IT Administrator	1.2	40 mins	Technical delivery operator 1
1.4	Make sure site is level	IT Manager	1.3	30 mins	Technical delivery operator 1

1.5	Unload equipment from Transport	Delivery van	1.4	4 hours	Technical delivery operator 1
1.6	Move it to install site		1.5	2 hours	Technical delivery operator 1
1.7	Insall anti vibration anchors		1.6	1 hour	Technical delivery operator 1
1.8	Ensure equipment controls are accessible		1.7	30 mins	Technical delivery operator 1
1.9	Ensure equipment controls compliance		1.8	30 mins	Technical delivery operator 1
1.10	Review fire safety regs		1.9	40 mins	Technical delivery operator 1

Action Plan for Work Package 2.0: Connect to plumbing					
Deliverables on completion of Work Package: Equipment connected to site plumbing					
Measures of Accomplishments / Milestones Turn off water supply Securely attach drain line					
Key constraints and Assumptions Assume the equipment has been placed and the site has been prepared					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
2.0	Connect to plumbing				
2.1	Check safety regs			1 day	Plumber 1
2.2	Check if additional parts are needed	Toolbox	2.1	1 day	Plumber 1
2.3	Turn off water supply		2.2	1 hour	Plumber 1
2.4	Locate nearest water supply & waste line		2.3	1 hour	Plumber 1
2.5	Install shutoff valve	Shutoff valve	2.4	1 hour	Plumber 1
2.6	Attach flexi hose to water inlet	Flexi hose	2.5	1 hour	Plumber 1

2.7	Use Teflon tape on threaded connections	Teflon tape	2.6	30 mins	Plumber 1
2.8	Install backflow preventer	Backflow preventer	2.7	40 mins	Plumber 1
2.9	Securely attach drain line		2.8	40 mins	Plumber 1
2.10	Check for leaks by slowly turning on water supply		2.9	40 mins	Plumber 1

Action Plan for Work Package 3.0: Connect to electricity					
Deliverables on completion of Work Package: Equipment connected to site electricity					
Measures of Accomplishments / Milestones Connect to dedicated circuit via conduit Turn power back on					
Key constraints and Assumptions Assume the equipment has been delivered and the site has been prepped					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
3.0	Connect to electricity				
3.1	Verify appliance specs against building panel capacity	Electricians' toolbox		1 day	Electrician 1
3.2	Turn off power to site		3.1	1 day	Electrician 1
3.3	Check if socket is accessible		3.2	1 hour	Electrician 1
3.4	If required, connect directly to power		3.3	1 hour	Electrician 1
3.5	Install disconnect switch	Disconnect switch	3.4	1 hour	Electrician 1
3.6	Connect to dedicated circuit via conduit	Electrical conduit	3.5	1 hour	Electrician 1
3.7	Attach grounding wire	Grounding wire	3.6	1 hour	Electrician 1

3.8	Verify connections are secure & compliant		3.7	1 hour	Electrician 1
3.9	Turn the circuit breaker back on		3.8	20 mins	Electrician 1
3.10	Test equipment to verify power		3.9	30 mins	Electrician 1

Action Plan for Work Package 4.0: Connect to network					
Deliverables on completion of Work Package: Equipment is connected to the site network					
Measures of Accomplishments / Milestones Remote monitoring is working					
Key constraints and Assumptions Assume the equipment has been delivered and the electricity has been connected					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
4.0	Connect to network				
4.1	Review equipment manual for network specs	IT technicians' toolbox		1 day	IT technician 1
4.2	Verify network availability		4.1	1 day	IT technician 1
4.3	Create dedicated VLAN on network		4.2	1 hour	IT technician 1
4.4	Input network credentials via control panel		4.3	1 hour	IT technician 1
4.5	If wired, use CAT6 link to router/switch	CAT6 cabling	4.4	1 hour	IT technician 1
4.6	Secure ethernet cabling	Ethernet cable clips	4.5	40 mins	IT technician 1
4.7	Run diagnostics to confirm integration		4.6	40 mins	IT technician 1
4.8	Enable remote monitoring		4.7	20 mins	IT technician 1
4.9	Set alerts and clear any errors		4.8	20 mins	IT technician 1
4.10	Verify user permissions		4.9	20 mins	IT technician 1

Action Plan for Work Package 5.0: Testing equipment					
Deliverables on completion of Work Package: Equipment has been tested and work documented					
Measures of Accomplishments / Milestones Equipment is undamaged and installed correctly Installation is documented Client is trained and up to knowledge on use of equipment					
Key constraints and Assumptions Assume equipment has been installed (Plumbing, electricity, network)					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
5.0	Testing equipment				
5.1	Perform visual inspection for damage			30 mins	QA inspector 1
5.2	Confirm equipment powers on		5.1	25 mins	QA inspector 1
5.3	Inspect plumbing for leaks		5.2	40 mins	QA inspector 1
5.4	Test equipment alarms and emergency shutoff		5.3	1 hour	QA inspector 1
5.5	Run temp/pressure test on equipment		5.4	1 hour	QA inspector 1
5.6	Confirm IoT functionality		5.5	1 hour	QA inspector 1
5.7	Adjust settings for client		5.6	1 hour	QA inspector 1
5.8	Document equipment setup and settings		5.7	1 hour	QA inspector 1
5.9	Provide client with user instructions and training		5.8	1 day	QA inspector 1
5.10	Schedule inspection/maintenance		5.9	25 mins	QA inspector 1

Action Plan for Work Package 6.0: Design and printing					
Deliverables on completion of Work Package: Marketing material designed and printed					
Measures of Accomplishments / Milestones Draft initial design Print brochures					
Key constraints and Assumptions Assume market research has been conducted					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible

6.0	Design and printing				
6.1	Identify target audience and message			1 day	Marketing assistant 1
6.2	Take insights from market research	Market research reports	6.1	1 day	Marketing assistant 1
6.3	Draft content for brochure	Digital design software	6.2	1 hour	Marketing assistant 1
6.4	Set design guidelines for branding		6.3	1 hour	Marketing assistant 1
6.5	Draft initial design		6.4	2 hours	Marketing assistant 1
6.6	Collect feedback on design and revise		6.5	2 hours	Marketing assistant 1
6.7	Proofread final version		6.6	1 hour	Marketing assistant 1
6.8	Prepare print files		6.7	1 hour	Marketing assistant 1
6.9	Select printing vendor		6.8	2 hours	Marketing assistant 1
6.10	Print brochures		6.9	2 hours	Marketing assistant 1

Action Plan for Work Package 7.0: Distribution					
Deliverables on completion of Work Package: Brochures distributed					
Measures of Accomplishments / Milestones Sort and package brochures for delivery Distribute at events/location					
Key constraints and Assumptions Assume market research has been conducted, brochures have been designed and printed, and campaign has been designed					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
7.0	Distribution				
7.1	Plan distribution channels (in-person, mail)			1 day	Marketing assistant 1
7.2	Define target audience	Market research reports	7.1	1 day	Marketing assistant 1

7.3	Sort and package brochures for delivery		7.2	2 hours	Marketing assistant 1
7.4	Arrange transportation or delivery services		7.3	1 hour	Marketing assistant 1
7.5	Distribute at events/location		7.4	2 hours	Marketing assistant 1
7.6	Use retailers and partners for brochure placement		7.5	2 hours	Marketing assistant 1
7.7	Track distribution progress		7.6	1 hour	Marketing assistant 1
7.8	Collect feedback		7.7	1 hour	Marketing assistant 1

Action Plan for Work Package 8.0: Campaign design					
Deliverables on completion of Work Package: Campaign media designed and planned					
Measures of Accomplishments / Milestones Draft initial design Print brochures					
Key constraints and Assumptions Assume market research has been conducted					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
8.0	Campaign design				
8.1	Define campaign objectives			1 day	Marketing assistant 1
8.2	Identify target audience from market research	Market research reports	8.1	1 day	Marketing assistant 1
8.3	Develop content from campaign goals	Digital design software	8.2	3 hours	Marketing assistant 1
8.4	Choose social media platforms		8.3	1 hour	Marketing assistant 1
8.5	Create graphics, videos, text for		8.4	2 hours	Marketing assistant 1

	campaign content				
8.6	Schedule posts and activities		8.5	2 hours	Marketing assistant 1
8.7	Adjust designs for platform specs		8.6	1 hour	Marketing assistant 1
8.8	Plan paid promotions for posts or ads		8.7	2 hours	Marketing assistant 1

Action Plan for Work Package 9.0: Booking/Reservation					
Deliverables on completion of Work Package: Booked and planned attendance at trade show					
Measures of Accomplishments / Milestones Choose size and location of booth Arrange for travel, accommodation, and materials					
Key constraints and Assumptions Assume campaign design has been conducted					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
9.0	Booking/Reservation				
9.1	Research relevant trade shows			1 day	Marketing assistant 1
9.2	Choose size and location of booth		9.1	2 hours	Marketing assistant 1
9.3	Confirm budget for fees and logistics		9.2	1 hour	Marketing assistant 1
9.4	Reserve the booth		9.3	1 hour	Marketing assistant 1
9.5	Arrange for travel, accommodation, and materials		9.4	1 day	Marketing assistant 1

Action Plan for Work Package 10.0: Surveys					
Deliverables on completion of Work Package: Conducted market research and market research report					
Measures of Accomplishments / Milestones Design concise questions based on goals Create insights from survey data					
Key constraints and Assumptions					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
10.0	Surveys				

10.1	Identify specific information to collect			2 hours	Marketing assistant 1
10.2	Design concise questions based on goals		10.1	2 hours	Marketing assistant 1
10.3	Determine demographics for survey		10.2	1 hour	Marketing assistant 1
10.4	Send out surveys through email, in-person, social media		10.3	1 day	Marketing assistant 1
10.5	Review data from responses		10.4	1 day	Marketing assistant 1
10.6	Create insights from survey data		10.5	1 day	Marketing assistant 1

5.8 action plans

Action Plan for Work Package 1.0: recruitment					
Deliverables on completion of Work Package: hire new employees.					
Measures of Accomplishments / Milestones: **Hire new people with correct skills and qualifications.**					
Key constraints and Assumptions					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
1.0	Hiring	Recruitment specialist			HR manager
1.1	Define project specified roles			1 day	HR officer
1.2	outline required skill sets	HR team, project lead	1.1	1 day	Hr analysist
1.3	develop job descriptions	HR team	1.2	1 hour	Hr analysist
1.4	post job openings	HR tools (LinkedIn, indeed)	1.3	1 hour	HR assistant
1.5	screen applications	Recruitment software	1.4	3 days	HR officer
1.6	conduct interviews	Interview panel	1.5	5 days	HR manager, Line Manager

1.7	assess candidates	HR team, evaluation forms	1.6	2 days	HR officer
1.8	negotiate employment terms	Officer letters, policies	1.7	2 days	HR manager
1.9	hire and onboard new people	Orientation materials	1.8	1 day	HR team

Action Plan for Work Package 2.0: safety					
Deliverables on completion of Work Package: safety training + compliances with safety standards + emergency preparation					
Measures of Accomplishments / Milestones Train staff with the right tools for safety					
Key constraints and Assumptions					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
2.0	Train staff in safety	Safety team, trainers, tools			Safety Manager
2.1	develop safety training materials for IOT system	Safety manuals, tools		1 day	Training Specialist
2.2	conduct workshops on equipment	Equipment, trainers	2.1	3 days	Safety Trainer
2.3	educate staff on safety risks	Workshop content	2.2	1 day	Safety Trainer
2.4	provide digital training modules for flexibilty	Learning Management System	2.3	1 day	Training Specialist
2.5	train employees on fire and electric protocols	Fire protocol manuals, trainers	2.4	2 days	Safety Officer
2.6	test employee knowledge through quizzes	LMS software	2.5	1 day	Training Specialist
2.7	offer regular health and safety check-ups	Medical team	2.6	1 day	Safety Manager
2.8	ensure all training aligns with OSHA	OSHA guidelines	2.7	2 days	Compliance Officer

2.9	collab with tech teams to meet codes	Tech teams	2.8	1 day	Safety Manager
2.10	document safety measures for audits	Safety compliance tools	2.9	1 day	Compliance Officer
2.11	update policies to reflect changing regs	Legal and Compliance Experts	2.10	3 days	Legal Advisor
2.12	partner with external consultants for safety certs	Certification Consultants	2.11	2 days	Compliance Officer
2.13	organize routine compliance audits	Internal Audit Teams	2.12	2 days	Compliance Officer
2.14	monitor vendor compliance	Vendor Reports, Monitoring Tools	2.13	2 days	Procurement Manager
2.15	conduct fire drills regularly	Fire Safety Equipment, Drills	2.14	1 day	Safety Trainer
2.16	train employees on evacuation routes	Evacuation Maps, Safety Trainers	2.15	1 day	Safety Trainer
2.17	Develop contingency plans for system failures	Contingency Experts, IT Team	2.16	3 days	Risk Management Officer
2.18	provide training on using fire extinguishers	Fire Extinguishers, Trainers	2.17	1 day	Safety Officer
2.19	partner with technical teams to test IOT emergency alerts	Alert Systems	2.18	2 days	Safety Manager

Action Plan for Work Package 3.0: employee wellbeing
Deliverables on completion of Work Package:
Measures of Accomplishments / Milestones

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Key constraints and Assumptions

Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
3.0	Ensure employee wellbeing	Wellbeing Team, Ergonomics Tools			Wellbeing Manager
3.1	provide ergonomic assessments	Ergonomic Experts, Tools		2 days	Health & Safety Specialist
3.2	ensure proper lighting and ventilation	Facilities Team, IoT Sensors	3.1	2 days	Facilities Manager
3.3	monitor stress and workload	Surveys, HR Tools	3.2	1 day	HR Specialist
3.4	encourage mental health recourses	Counselling Services, Apps	3.3	2 days	HR Specialist
3.5	provide access to first-aid stations	Medical Supplies, Facilities	3.4	1 day	Facilities Manager
3.6	implement IOT-driven tools to track fatigue	IoT Health Tools	3.5	3 days	Wellbeing manager
3.7	offer regular health and safety check-ups	Medical Team	3.6	1 day (Monthly)	Health & Safety Specialist
3.8	satisfaction surveys	Survey Tools, HR Platforms	3.7	1 day	HR Specialist
3.9	develops an incident reporting system	HR Systems, IT Team	3.8	2 days	IT Manager, HR Specialist
3.10	train employees on reporting procedures	Training Modules, HR Trainers	3.9	1 day	HR Trainer
3.11	investigate safety violations	Internal Auditors, Legal Advisors	3.10	2 days	Health & Safety Specialist
3.12	set up emergency hotline	IT Support, Communication Tool	3.11	1 day	IT Manager
3.13	create clear policies for handling workplace injuries	Legal Team, HR Policies	3.12	3 days	HR Manager, Legal Advisor

3.14	document roots causes and corrective actions	Incident Reports, HR Tools	3.13	2 days	Health & Safety Specialist

Action Plan for Work Package 4.0: training + development					
Deliverables on completion of Work Package: create a training package for employees and train them.					
Measures of Accomplishments / Milestones …					
Key constraints and Assumptions					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
4.0	Make training	Training Team, HR Resources			Training Manager
4.1	analyse organizational skills goals + identify training needs	HR Analytics Tools, Managers		2 days	HR Analyst
4.2	conduct surveys or interviews with employees to gather input	Survey Tools, Interview Guides	4.1	2 days	HR Specialist
4.3	review performance evaluations	HR Systems	4.2	1 day	HR Specialist
4.4	collaborate with department heads to understand specific teams needs	Meetings, Communication Tools	4.3	2 days	Training Manager
4.5	determine the priority of training programs	Prioritization Frameworks	4.4	2 days	Training Manager
4.6	set clear training objectives aligned with business goals	Organizational Goals, Team Input	4.5	1 day	Training Manager
4.7	create a budget for training resources	Budgeting Tools	4.6	2 days	Finance Officer
4.8	define success metrics for training	Analytics Tools, KPIs	4.7	1 day	HR Analyst

4.9	develop training materials	Content Creation Software	4.8	5 days	Instructional Designer
4.10	select training methods	Research, Surveys	4.9	2 days	Training Manager
4.11	coordinate the recruitment	Recruitment Tools	4.10	3 days	HR Specialist
4.12	prepare coordination for training venues	Venue Booking Systems, IT Support	4.11	2 days	Facilities Manager
4.13	create a registration process	Software for Registration	4.12	1 day	IT Manager
4.14	deliver the training sessions	Trainers, Training Materials	4.13	10 days	Trainers
4.15	gather info from participants	Surveys, Feedback Forms	4.14	1 day	HR Specialist
4.16	assess their knowledge	Quizzes, Testing Tools	4.15	2 days	Trainers
4.17	analyse feedback	Data Analysis Tools, Feedback Forms	4.16	2 days	Training Manager

Action Plan for Work Package 5.0: system integration					
Deliverables on completion of Work Package: create a positive workspace.					
Measures of Accomplishments / Milestones …					
Key constraints and Assumptions					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
5.0	Change the system	HR Team, IT Support, Trainers			Change Manager
5.1	change management and communication	Change Management Tools		3 days	Change Manager
5.2	communicate the benefits of new system	Presentations, Email Tools	5.1	2 days	HR Communications Specialist
5.3	design communication materials	Graphic Designers, Software	5.2	2 days	Marketing Team
5.4	organize info sessions	Scheduling Tools, Venues	5.3	2 days	HR Specialist
5.5	address concerns and resistant	Feedback Tools, Meetings	5.4	3 days	Change Manager

5.6	assign HR reps	HR Team	5.5	2 days	HR Manager
5.7	develop a plan for ongoing comms	Email Tools, Social Platforms	5.6	2 days	HR Communications Specialist
5.8	ensure clarity around timelines	Project Management Tools	5.7	2 days	Project Manager
5.9	develop training content specific to new system	Learning Management System (LMS)	5.8	5 days	Instructional Designer
5.10	create user guides, manuals	Content Creation Tools	5.9	3 days	Instructional Designer
5.11	organize workshops	Trainers, Venues	5.10	5 days	Training Manager
5.12	coordinate one-on-one trainning	Trainers, Scheduling Tools	5.11	5 days	HR Specialist, Trainers
5.13	ensure trainers are well-versed	Training for Trainers, Materials	5.12	2 days	Training Manager
5.14	set up internal helpdesk	IT Support Staff, Software	5.13	2 days	IT Manager
5.15	provide ongoing assistance/troubleshooting	Helpdesk Tools, IT Staff	5.14	5 days	IT Support Team

Action Plan for Work Package 6.0: project management					
Deliverables on completion of Work Package:					
Measures of Accomplishments / Milestones …					
Key constraints and Assumptions					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
6.0	Creating project teams	HR Team, Leadership Team			Project Manager
6.1	identify staffing requirements for project team	Staffing Analysis Tools		3 days	HR Manager
6.2	asses current employee capabilities and potential	Skills Assessment Tools	6.1	3 days	HR Specialist

6.3	create job descriptions for project-specific roles	Content Creation Tools	6.2	2 days	HR Specialist
6.4	collab with hiring managers to recruit internal project team members	Interview Panels, Talent Pool Database	6.3	3 days	Recruitment Manager
6.5	ensure project teams have a balance or required skills	Competency Matrices	6.4	2 days	Project Manager
6.6	onboard new hires employees	Orientation Materials, IT Tools	6.5	3 days	HR Specialist
6.7	develop a timeline for employee assignments	Project Management Software	6.6	2 days	Project Manager
6.8	review team strucutres	Collaboration Tools	6.7	3 days	HR Specialist
6.9	monitor project workload	Resource Allocation Software	6.8	3 days	Project Manager
6.10	set clear performance expectations	Goal-Setting Frameworks	6.9	2 days	Team Leads
6.11	provide feedback and coaching	Communication Tools	6.10	5 days	Team Leads
6.12	address performance issues and create devolpment plans	HR Policy Guidelines	6.11	3 days	HR Manager
6.13	facilitate mentoring and coaching programs	Mentorship Frameworks	6.12	5 days	HR Specialist
6.14	encourage proffesional devolpmnet	Learning Platforms, Workshops	6.13	5 days	Training Manager
6.15	recognize and reward high prefomers	Incentive Programs, Tools for Tracking	6.14	3 days	HR Manager

Action Plan for Work Package 7.0: market research
Deliverables on completion of Work Package:
Measures of Accomplishments / Milestones

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Key constraints and Assumptions

Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
7.0	Do market research	Market Analysis Tools, Surveys			Marketing Manager
7.1	Conduct surveys and focus groups with target consumers	Survey Platforms, Participant Database		3 days	Market Research Analyst
7.2	Analyse competitor products and strategies	Competitor Reports, SWOT Analysis	7.1	3 days	Market Research Analyst
7.3	Utilize market intelligence tools to gather data	Market Intelligence Software	7.2	2 days	Data Analyst
7.4	Identify potential opportunities for innovation	Market Insights, Brainstorming Tools	7.3	2 days	Product Manager
7.5	Create detailed reports on market demand and gaps	Reporting Software, Templates	7.4	3 days	Data Analyst
7.6	Study customer feedback and pain points	Feedback Forms, Customer Data	7.5	3 days	Market Research Analyst
7.7	Evaluate usage patterns for related products	Analytics Software	7.6	2 days	Data Analyst
7.8	Engage with key stakeholders to define requirements	Stakeholder Engagement Tools	7.7	3 days	Product Manager
7.9	Partner with marketing teams to refine consumer personas	Marketing Data, Focus Group Results	7.8	2 days	Marketing Specialist
7.10	Assess financial viability for product concepts	Financial Analysis Software	7.9	4 days	Financial Analyst
7.11	Conduct cost-benefit analysis	Costing Models, Scenario Planning Tools	7.10	3 days	Financial Analyst

Action Plan for Work Package 8.0: technology innovation

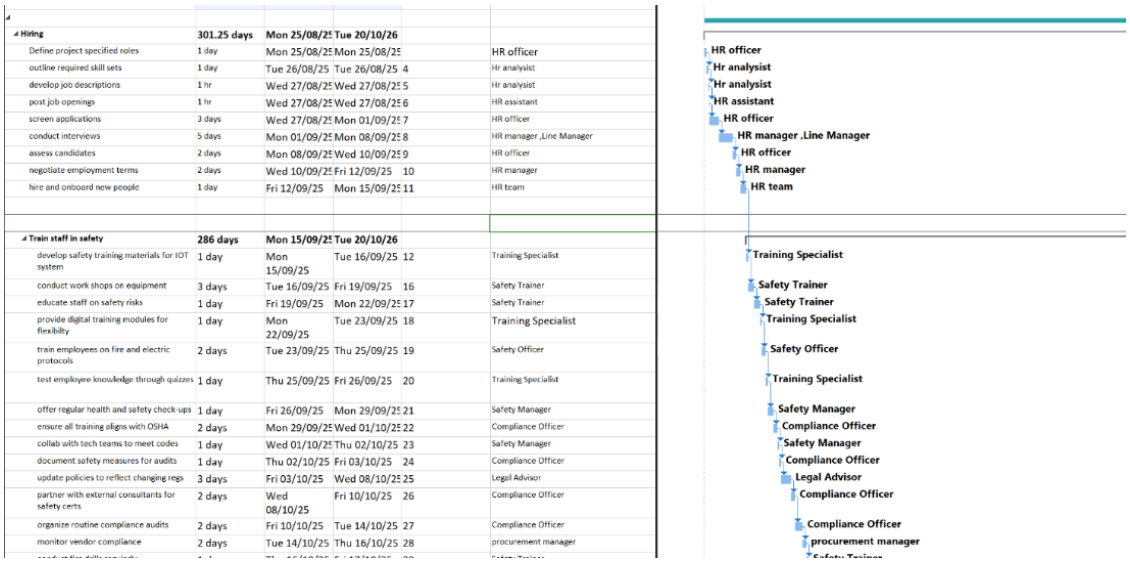
Deliverables on completion of Work Package:

Measures of Accomplishments / Milestones …					
Key constraints and Assumptions					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
8.0	Brainstorm for tech innovation	Collaboration Tools, Idea Management			Innovation Manager
8.1	Host brainstorming sessions and innovation workshops	Meeting Rooms, Facilitation Tools		2 days	Innovation Specialist
8.2	Encourage cross-functional collaboration for diverse insights	Collaboration Platforms	8.1	3 days	Innovation Specialist
8.3	Partner with external tech incubators and start-ups	Networking Tools, External Consultants	8.2	3 days	Product Manager
8.4	Leverage AI and big data to generate insights	AI Software, Data Platforms	8.3	2 days	Data Scientist
8.5	Explore patent landscapes to inspire unique solutions	Patent Databases	8.4	2 days	Legal Advisor
8.6	Design blueprints and initial schematics	CAD Tools, Design Software	8.5	5 days	Engineering Team
8.7	Test viability through simulations and models	Simulation Tools, Testing Software	8.6	5 days	Engineering Team
8.8	Collaborate with engineering teams for technical inputs	Collaboration Platforms	8.7	2 days	Innovation Manager
8.9	Identify potential technologies to integrate	Market Reports, Feasibility Studies	8.8	3 days	Product Manager
8.10	Assess scalability of proposed ideas	Prototyping Tools	8.9	3 days	Engineering Team

Action Plan for Work Package 9.0: prototyping and product testing					
Deliverables on completion of Work Package:					
Measures of Accomplishments / Milestones …					
Key constraints and Assumptions					
Number	Task	Resources	Preceding task	Duration	Assigned to / Responsible
9.0	Begin prototyping and testing	Prototyping Tools			Product Manager
9.1	Build functional prototypes based on approved concepts	3D Printers, Engineering software		5 days	Engineering Team
9.2	Use rapid prototyping tool	3D Printers, Test Equipment	9.1	5 days	Engineering Team
9.3	Test prototypes for usability and performance	Usability Testing Labs, Feedback Tools	9.2	5 days	QA Specialist
9.4	Refine designs based on stakeholder feedback	Design Software, CAD Tools	9.3	5 days	Engineering Team
9.5	Document prototype iterations for future reference	Documentation Too	9.4	2 days	Technical Writer
9.6	Conduct lab-based testing for durability and efficiency	Lab Equipment	9.5	5 days	QA Specialist
9.7	Organize focus groups for user testing and feedback	Participants	9.6	5 days	Marketing Manager
9.8	Partner with QA teams for rigorous testing protocols	Survey Tools	9.7	3 days	Market Research Analyst
9.9	Monitor field testing for real-world performance data	Presentation Tool	9.8	2 days	Product Manager
9.10	Compare results against defined success criteria	Prototyping Tools	9.9	5 days	Engineering Team

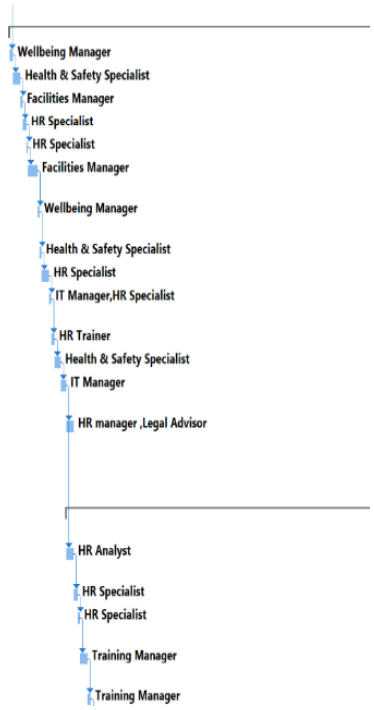
6.0 Project Scheduling

HR:



Hiring	301.25 days	Mon 25/08/25	Tue 20/10/26		
Define project specified roles	1 day	Mon 25/08/25	Mon 25/08/25		HR officer
outline required skill sets	1 day	Tue 26/08/25	Tue 26/08/25	4	Hr analysist
develop job descriptions	1 hr	Wed 27/08/25	Wed 27/08/25	5	Hr analysist
post job openings	1 hr	Wed 27/08/25	Wed 27/08/25	6	HR assistant
screen applications	3 days	Wed 27/08/25	Mon 01/09/25	7	HR officer
conduct interviews	5 days	Mon 01/09/25	Mon 08/09/25	8	HR manager ,Line Manager
assess candidates	2 days	Mon 08/09/25	Wed 10/09/25	9	HR officer
negotiate employment terms	2 days	Wed 10/09/25	Fri 12/09/25	10	HR manager
hire and onboard new people	1 day	Fri 12/09/25	Mon 15/09/25	11	HR team
Train staff in safety	286 days	Mon 15/09/25	Tue 20/10/26		
develop safety training materials for IOT system	1 day	Mon 15/09/25	Tue 16/09/25	12	Training Specialist
conduct work shops on equipment	3 days	Tue 16/09/25	Fri 19/09/25	16	Safety Trainer
educate staff on safety risks	1 day	Fri 19/09/25	Mon 22/09/25	17	Safety Trainer
provide digital training modules for flexibilty	1 day	Mon 22/09/25	Tue 23/09/25	18	Training Specialist
train employees on fire and electric protocols	2 days	Tue 23/09/25	Thu 25/09/25	19	Safety Officer
test employee knowledge through quizzes	1 day	Thu 25/09/25	Fri 26/09/25	20	Training Specialist
offer regular health and safety check-ups	1 day	Fri 26/09/25	Mon 29/09/25	21	Safety Manager
ensure all training aligns with OSHA	2 days	Mon 29/09/25	Wed 01/10/25	22	Compliance Officer
collab with tech teams to meet codes	1 day	Wed 01/10/25	Thu 02/10/25	23	Safety Manager
document safety measures for audits	1 day	Thu 02/10/25	Fri 03/10/25	24	Compliance Officer
update policies to reflect changing regs	3 days	Fri 03/10/25	Wed 08/10/25	25	Legal Advisor
partner with external consultants for safety certs	2 days	Wed 08/10/25	Fri 10/10/25	26	Compliance Officer
organize routine compliance audits	2 days	Fri 10/10/25	Tue 14/10/25	27	Compliance Officer
monitor vendor compliance	2 days	Tue 14/10/25	Thu 16/10/25	28	procurement manager

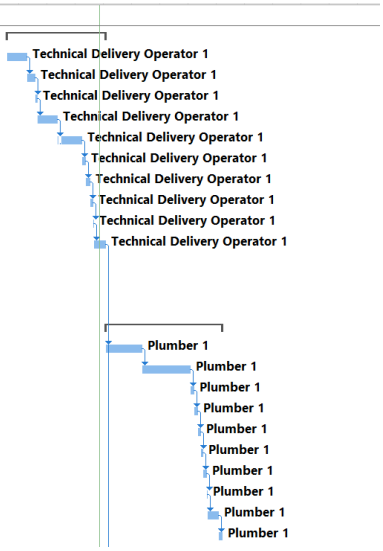
Ensure employee wellbeing	**255 days**	**Tue 28/10/25**	**Tue 20/10/26**		
provide ergonomic assessments	2 days	Tue 28/10/25	Thu 30/10/25	34	Wellbeing Manager
ensure proper lighting and ventilation	2 days	Thu 30/10/25	Mon 03/11/25	38	Health & Safety Specialist
monitor stress and workload	1 day	Mon 03/11/25	Tue 04/11/25	39	Facilities Manager
encourage mental health recourses	2 days	Tue 04/11/25	Thu 06/11/25	40	HR Specialist
provide access to first-aid stations	1 day	Thu 06/11/25	Fri 07/11/25	41	HR Specialist
implement IOT-driven tools to track fatigue	3 days	Fri 07/11/25	Wed 12/11/25	42	Facilities Manager
offer regular health and safety check-ups	1 day	Wed 12/11/25	Thu 13/11/25	43	Wellbeing Manager
satisfaction surveys	1 day	Thu 13/11/25	Fri 14/11/25	44	Health & Safety Specialist
devolp an incident reporting systems	2 days	Fri 14/11/25	Tue 18/11/25	45	HR Specialist
train employees on reporting procedures	1 day	Tue 18/11/25	Wed 19/11/25	46	IT Manager,HR Specialist
investigate safety violations	2 days	Wed 19/11/25	Fri 21/11/25	47	HR Trainer
set up emergency hotline	1 day	Fri 21/11/25	Mon 24/11/25	48	Health & Safety Specialist
create clear policies for handling workplace injuires	3 days	Mon 24/11/25	Thu 27/11/25	49	IT Manager
document roots causes and corrective actions	2 days	Thu 27/11/25	Mon 01/12/25	50	HR manager ,Legal Advisor
analyze organizational skills goals + identify training needs	**233 days**	**Thu 27/11/25**	**Tue 20/10/26**		
conduct surveys or interviews with employees to gather input	2 days	Thu 27/11/25	Mon 01/12/25	50	HR Analyst
review performance evaluations	2 days	Mon 01/12/25	Wed 03/12/25	55	HR Specialist
collabrate with department heads to understand specific teams needs	1 day	Wed 03/12/25	Thu 04/12/25	56	HR Specialist
determine the priority of training programs	2 days	Thu 04/12/25	Mon 08/12/25	57	Training Manager
set clear training objectives aligned	2 days	Mon	Wed	58	Training Manager

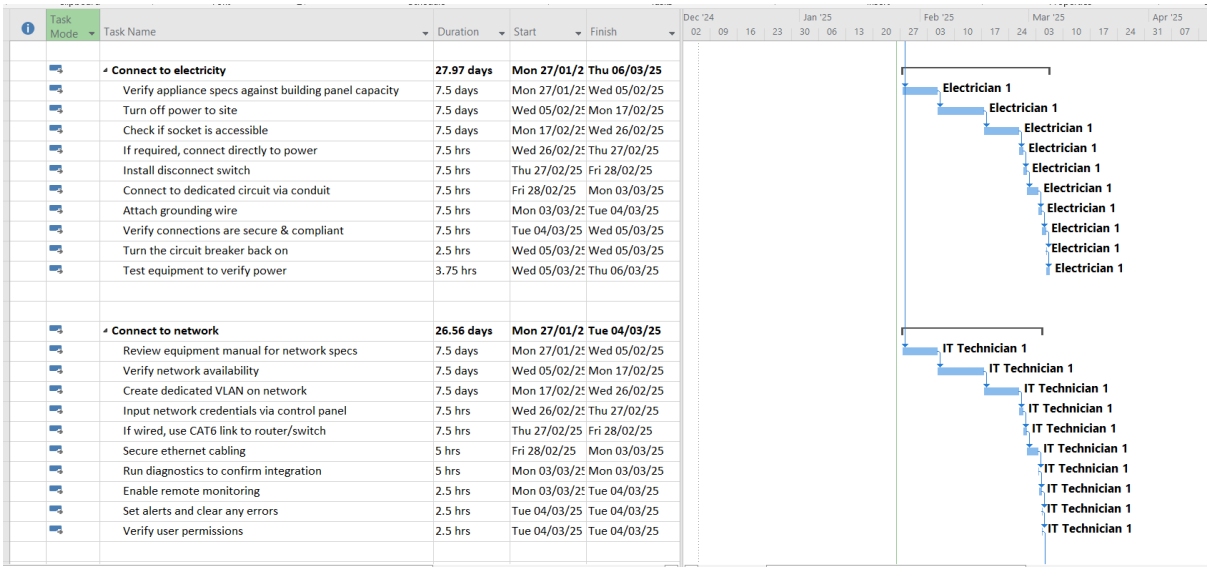


R&D:

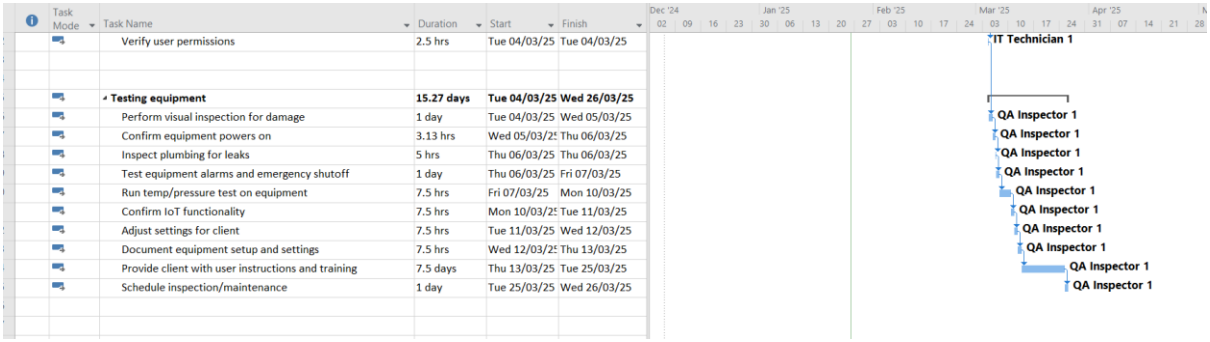
Technical Installation:

		Equipment Placement	**16.19 days**	**Fri 03/01/25**	**Mon 27/01/25**
		Clear installation site	3 days	Fri 03/01/25	Tue 07/01/25
		Measure site dimensions	2 days	Wed 08/01/25	Thu 09/01/25
		Check for utilities service	5 hrs	Fri 10/01/25	Fri 10/01/25
		Make sure site is level	3 days	Fri 10/01/25	Wed 15/01/25
		Unload equipment from Transport	30 hrs	Wed 15/01/25	Tue 21/01/25
		Move it to install site	1 day	Tue 21/01/25	Wed 22/01/25
		Insall anti vibration anchors	1 day	Wed 22/01/25	Thu 23/01/25
		Ensure equipment controls are accessible	3.75 hrs	Thu 23/01/25	Fri 24/01/25
		Ensure equipment controls compliance	3.75 hrs	Fri 24/01/25	Fri 24/01/25
		Review fire safety regs	5 hrs	Fri 24/01/25	Mon 27/01/25
		Connect to plumbing	**20.95 days**	**Mon 27/01/2**	**Tue 25/02/25**
		Check safety regs	7.5 days	Mon 27/01/25	Wed 05/02/25
		Check if additional parts are needed	7.5 days	Wed 05/02/25	Mon 17/02/25
		Turn off water supply	7.5 hrs	Mon 17/02/25	Tue 18/02/25
		Locate nearest water supply & waste line	7.5 hrs	Tue 18/02/25	Wed 19/02/25
		Install shutoff valve	7.5 hrs	Wed 19/02/25	Wed 19/02/25
		Attach flexi hose to water inlet	7.5 hrs	Thu 20/02/25	Thu 20/02/25
		Use Teflon tape on threaded connections	3.8 hrs	Thu 20/02/25	Fri 21/02/25
		Install backflow preventer	3.8 hrs	Fri 21/02/25	Fri 21/02/25
		Securely attach drain line	5 hrs	Fri 21/02/25	Mon 24/02/25
		Check for leaks by slowly turning on water supply	5 hrs	Mon 24/02/25	Tue 25/02/25



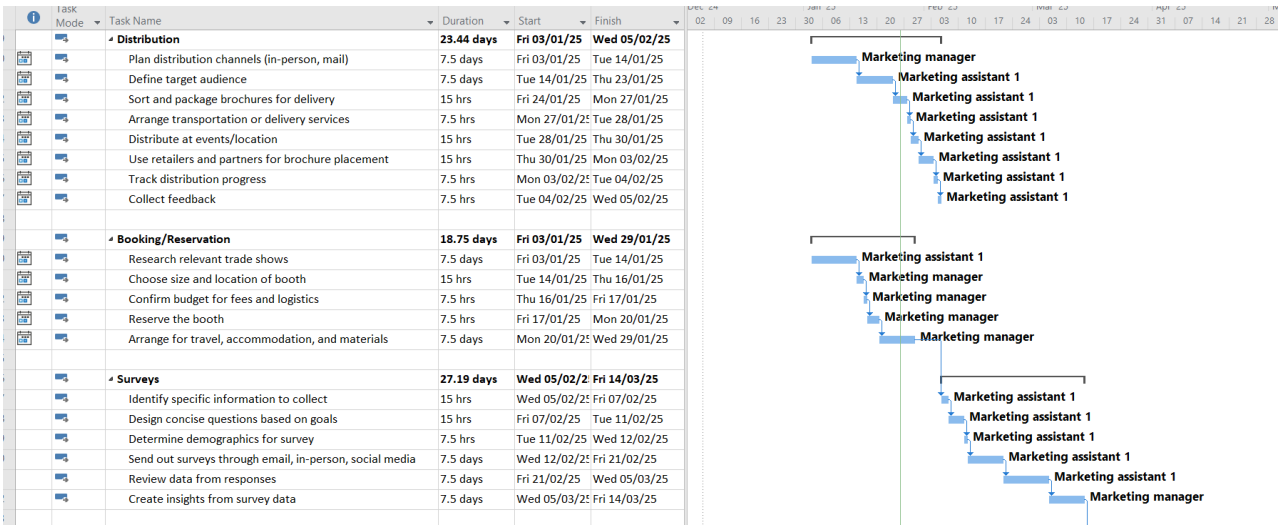


Task Mode	Task Name	Duration	Start	Finish
	Connect to electricity	**27.97 days**	**Mon 27/01/2**	**Thu 06/03/25**
	Verify appliance specs against building panel capacity	7.5 days	Mon 27/01/25	Wed 05/02/25
	Turn off power to site	7.5 days	Wed 05/02/25	Mon 17/02/25
	Check if socket is accessible	7.5 days	Mon 17/02/25	Wed 26/02/25
	If required, connect directly to power	7.5 hrs	Wed 26/02/25	Thu 27/02/25
	Install disconnect switch	7.5 hrs	Thu 27/02/25	Fri 28/02/25
	Connect to dedicated circuit via conduit	7.5 hrs	Fri 28/02/25	Mon 03/03/25
	Attach grounding wire	7.5 hrs	Mon 03/03/25	Tue 04/03/25
	Verify connections are secure & compliant	7.5 hrs	Tue 04/03/25	Wed 05/03/25
	Turn the circuit breaker back on	2.5 hrs	Wed 05/03/25	Wed 05/03/25
	Test equipment to verify power	3.75 hrs	Wed 05/03/25	Thu 06/03/25
	Connect to network	**26.56 days**	**Mon 27/01/2**	**Tue 04/03/25**
	Review equipment manual for network specs	7.5 days	Mon 27/01/25	Wed 05/02/25
	Verify network availability	7.5 days	Wed 05/02/25	Mon 17/02/25
	Create dedicated VLAN on network	7.5 days	Mon 17/02/25	Wed 26/02/25
	Input network credentials via control panel	7.5 hrs	Wed 26/02/25	Thu 27/02/25
	If wired, use CAT6 link to router/switch	7.5 hrs	Thu 27/02/25	Fri 28/02/25
	Secure ethernet cabling	5 hrs	Fri 28/02/25	Mon 03/03/25
	Run diagnostics to confirm integration	5 hrs	Mon 03/03/25	Mon 03/03/25
	Enable remote monitoring	2.5 hrs	Mon 03/03/25	Tue 04/03/25
	Set alerts and clear any errors	2.5 hrs	Tue 04/03/25	Tue 04/03/25
	Verify user permissions	2.5 hrs	Tue 04/03/25	Tue 04/03/25



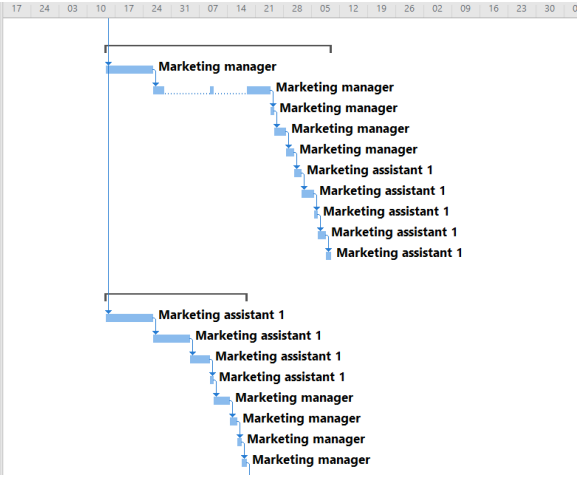
Task Mode	Task Name	Duration	Start	Finish
	Verify user permissions	2.5 hrs	Tue 04/03/25	Tue 04/03/25
	Testing equipment	**15.27 days**	**Tue 04/03/25**	**Wed 26/03/25**
	Perform visual inspection for damage	1 day	Tue 04/03/25	Wed 05/03/25
	Confirm equipment powers on	3.13 hrs	Wed 05/03/25	Thu 06/03/25
	Inspect plumbing for leaks	5 hrs	Thu 06/03/25	Thu 06/03/25
	Test equipment alarms and emergency shutoff	1 day	Thu 06/03/25	Fri 07/03/25
	Run temp/pressure test on equipment	7.5 hrs	Fri 07/03/25	Mon 10/03/25
	Confirm IoT functionality	7.5 hrs	Mon 10/03/25	Tue 11/03/25
	Adjust settings for client	7.5 hrs	Tue 11/03/25	Wed 12/03/25
	Document equipment setup and settings	7.5 hrs	Wed 12/03/25	Thu 13/03/25
	Provide client with user instructions and training	7.5 days	Thu 13/03/25	Tue 25/03/25
	Schedule inspection/maintenance	1 day	Tue 25/03/25	Wed 26/03/25

Marketing:



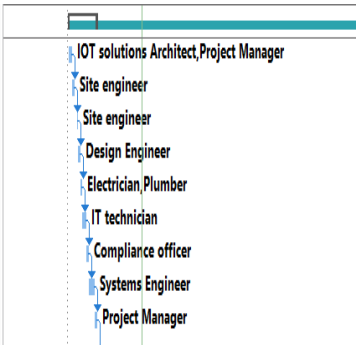
Task Mode	Task Name	Duration	Start	Finish
	Distribution	**23.44 days**	**Fri 03/01/25**	**Wed 05/02/25**
	Plan distribution channels (in-person, mail)	7.5 days	Fri 03/01/25	Tue 14/01/25
	Define target audience	7.5 days	Tue 14/01/25	Thu 23/01/25
	Sort and package brochures for delivery	15 hrs	Fri 24/01/25	Mon 27/01/25
	Arrange transportation or delivery services	7.5 hrs	Mon 27/01/25	Tue 28/01/25
	Distribute at events/location	15 hrs	Tue 28/01/25	Thu 30/01/25
	Use retailers and partners for brochure placement	15 hrs	Thu 30/01/25	Mon 03/02/25
	Track distribution progress	7.5 hrs	Mon 03/02/25	Tue 04/02/25
	Collect feedback	7.5 hrs	Tue 04/02/25	Wed 05/02/25
	Booking/Reservation	**18.75 days**	**Fri 03/01/25**	**Wed 29/01/25**
	Research relevant trade shows	7.5 days	Fri 03/01/25	Tue 14/01/25
	Choose size and location of booth	15 hrs	Tue 14/01/25	Thu 16/01/25
	Confirm budget for fees and logistics	7.5 hrs	Thu 16/01/25	Fri 17/01/25
	Reserve the booth	7.5 hrs	Fri 17/01/25	Mon 20/01/25
	Arrange for travel, accommodation, and materials	7.5 days	Mon 20/01/25	Wed 29/01/25
	Surveys	**27.19 days**	**Wed 05/02/2**	**Fri 14/03/25**
	Identify specific information to collect	15 hrs	Wed 05/02/25	Fri 07/02/25
	Design concise questions based on goals	15 hrs	Fri 07/02/25	Tue 11/02/25
	Determine demographics for survey	7.5 hrs	Tue 11/02/25	Wed 12/02/25
	Send out surveys through email, in-person, social media	7.5 days	Wed 12/02/25	Fri 21/02/25
	Review data from responses	7.5 days	Fri 21/02/25	Wed 05/03/25
	Create insights from survey data	7.5 days	Wed 05/03/25	Fri 14/03/25

Mode	Task Name	Duration	Start	Finish
	Design and Printing	**40.31 days**	**Fri 14/03/25**	**Fri 09/05/25**
	Identify target audience and message	7.5 days	Fri 14/03/25	Wed 26/03/25
	Take insights from market research	7.5 days	Wed 26/03/25	Thu 24/04/25
	Draft content for brochure	7.5 hrs	Thu 24/04/25	Fri 25/04/25
	Set design guidelines for branding	7.5 hrs	Fri 25/04/25	Mon 28/04/25
	Draft initial design	15 hrs	Mon 28/04/25	Wed 30/04/25
	Collect feedback on design and revise	15 hrs	Wed 30/04/25	Fri 02/05/25
	Proofread final version	7.5 hrs	Fri 02/05/25	Mon 05/05/25
	Prepare print files	7.5 hrs	Mon 05/05/25	Tue 06/05/25
	Select printing vendor	15 hrs	Tue 06/05/25	Thu 08/05/25
	Print brochures	15 hrs	Thu 08/05/25	Fri 09/05/25
	Campaign design	**25.31 days**	**Fri 14/03/25**	**Fri 18/04/25**
	Define campaign objectives	7.5 days	Fri 14/03/25	Wed 26/03/25
	Identify target audience from market research	7.5 days	Wed 26/03/25	Fri 04/04/25
	Develop content from campaign goals	21.5 hrs	Fri 04/04/25	Wed 09/04/25
	Choose social media platforms	7.5 hrs	Wed 09/04/25	Thu 10/04/25
	Create graphics, videos, text for campaign content	15 hrs	Thu 10/04/25	Mon 14/04/25
	Schedule posts and activities	15 hrs	Mon 14/04/25	Wed 16/04/25
	Adjust designs for platform specs	7.5 hrs	Wed 16/04/25	Thu 17/04/25
	Plan paid promotions for posts or ads	15 hrs	Thu 17/04/25	Fri 18/04/25



Project Support:

Site Assessment	18 days?	Mon 25/11/24	Wed 18/12/24		
Understand Requirements	3 days	Mon 25/11/24	Wed 27/11/24		IOT solutions Ar
Evaluate existing space	2 days	Thu 28/11/24	Fri 29/11/24	4	Site engineer
Check the layout	1 day	Mon 02/12/24	Mon 02/12/24	5	Site engineer
Evaluate environment	2 days	Tue 03/12/24	Wed 04/12/24	6	Design Engineer
Check utilities	1 day	Thu 05/12/24	Thu 05/12/24	7	Electrician,Plum
Test Connectivity	2 days	Fri 06/12/24	Mon 09/12/24	8	IT technician
Compliance Review	2 days	Tue 10/12/24	Wed 11/12/24	9	Compliance offi
Integration with existing systems	3 days	Thu 12/12/24	Mon 16/12/24	10	Systems Engine
Document plan	2 days	Tue 17/12/24	Wed 18/12/24	11	Project Manager



Project Manufacturing:

Logistics Management	18 days?	Mon 31/03/25	Wed 23/04/25		
Supply Chain Assessment	3 days	Mon 31/03/25	Wed 02/04/25	68	Logistics Manage
Vendor Coordination	2 days	Thu 03/04/25	Fri 04/04/25	71	Procurement Sp
Means of Transport	1 day	Mon 07/04/25	Mon 07/04/25	72	Transport coordi
Warehouse Management	2 days	Tue 08/04/25	Wed 09/04/25	73	Warehouse Man
Material Training	4 days	Thu 10/04/25	Tue 15/04/25	74	Warehouse supe
Production line Management	2 days	Wed 16/04/25	Thu 17/04/25	75	Logistics Manage
Quality assurance	3 days	Fri 18/04/25	Tue 22/04/25	76	Warehouse Man
Distribution / delivery	1 day	Wed 23/04/25	Wed 23/04/25	77	Contracted drive



7.0 Allocating Resources, Equipment and Personnel

Resource Name	Type	Material	Initials	Group	Max.	Std. Rate	Ovt. Rate	Cost/Use	Accrue	Base Calendar
HR manager	Work		H		100%	€30.77/hr	€46.16/hr	€0.00	Prorated	Standard
HR officer	Work		H		100%	€20.51/hr	€30.77/hr	€0.00	Prorated	Standard
HR assistant	Work		H		100%	€18.03/hr	€27.05/hr	€0.00	Prorated	Standard
Safety Manager	Work		S		100%	€33.33/hr	€50.00/hr	€0.00	Prorated	Standard
Safety Trainer	Work		S		100%	€16.92/hr	€25.38/hr	€0.00	Prorated	Standard
Training Specialist	Work		T		100%	€28.85/hr	€43.28/hr	€0.00	Prorated	Standard
Safety Officer	Work		S		100%	€25.64/hr	€38.46/hr	€0.00	Prorated	Standard
Compliance Officer	Work		C		100%	€29.49/hr	€44.24/hr	€0.00	Prorated	Standard
Legal Advisor	Work		L		100%	€40.53/hr	€60.80/hr	€0.00	Prorated	Standard
procurement manager	Work		p		100%	€37.18/hr	€55.77/hr	€0.00	Prorated	Standard
Risk Management Officer	Work		R		100%	€30.06/hr	€45.09/hr	€0.00	Prorated	Standard
Wellbeing Manager	Work		W		100%	€30.04/hr	€45.06/hr	€0.00	Prorated	Standard
Facilities Manager	Work		F		100%	€30.77/hr	€46.16/hr	€0.00	Prorated	Standard
HR Specialist	Work		H		100%	€24.36/hr	€36.54/hr	€0.00	Prorated	Standard
Health & Safety Specialist	Work		H		100%	€28.21/hr	€42.32/hr	€0.00	Prorated	Standard
IT Manager	Work		I		100%	€37.19/hr	€55.79/hr	€0.00	Prorated	Standard
HR Trainer	Work		H		100%	€16.00/hr	€24.00/hr	€0.00	Prorated	Standard
Training Manager	Work		T		100%	€27.72/hr	€41.58/hr	€0.00	Prorated	Standard
Finance Officer	Work		F		100%	€21.54/hr	€32.31/hr	€0.00	Prorated	Standard
HR Analyst	Work		H		100%	€16.77/hr	€25.16/hr	€0.00	Prorated	Standard
Instructional Designer	Work		I		100%	€28.85/hr	€43.28/hr	€0.00	Prorated	Standard
Trainers	Work		T		100%	€14.56/hr	€21.84/hr	€0.00	Prorated	Standard
HR Communications Specialist	Work		H		100%	€36.54/hr	€54.81/hr	€0.00	Prorated	Standard
Marketing Team	Work		M		100%	€18.72/hr	€28.08/hr	€0.00	Prorated	Standard
Change Manager	Work		C		100%	€38.38/hr	€57.57/hr	€0.00	Prorated	Standard
Project Manager	Work		P		100%	€37.03/hr	€55.55/hr	€0.00	Prorated	Standard
IT Support Team	Work		I		100%	€28.21/hr	€42.32/hr	€0.00	Prorated	Standard
Recruitment Manager	Work		R		100%	€26.92/hr	€40.38/hr	€0.00	Prorated	Standard
Team Leads	Work		T		100%	€30.06/hr	€45.09/hr	€0.00	Prorated	Standard
Marketing Manager	Work		M		100%	€28.21/hr	€42.32/hr	€0.00	Prorated	Standard
Market Research	Work		M		100%	€23.72/hr	€35.58/hr	€0.00	Prorated	Standard

Figure 7.1 HR Department Staff Allocation Table

Data Scientist	Work		D		100%	€34.23/hr	€51.35/hr	€0.00	Prorated	Standard
QA Specialist	Work		Q		100%	€28.21/hr	€42.32/hr	€0.00	Prorated	Standard
Technical Writer	Work		T		100%	€28.21/hr	€42.32/hr	€0.00	Prorated	Standard
Engineering Team	Work		E		100%	€43.59/hr	€65.39/hr	€0.00	Prorated	Standard
Line Manager	Work		L		100%	€20.85/hr	€57.26/hr	€0.00	Prorated	Standard
HR team	Work		H		100%	€17.16/hr	€51.35/hr	€0.00	Prorated	Standard
Procurement	Work		P		100%	€30.58/hr	€42.32/hr	€0.00	Prorated	Standard
Manager	Work		M		100%	€27.80/hr	€41.70/hr	€0.00	Prorated	Standard
Innovation Specialist	Work		I		100%	€29.66/hr	€44.49/hr	€0.00	Prorated	Standard
recruitment software	Material		r			€250.00		€0.00	Prorated	
HR manager platform	Material		H			€200.00		€0.00	Prorated	
interview software	Material		i			€250.00		€0.00	Prorated	
audit software	Material		a			€1,000.00		€0.00	Prorated	
learning management system	Material		l			€500.00		€0.00	Prorated	
medial supply	Material		m			€300.00		€0.00	Prorated	
survey tools	Material		s			€50.00		€0.00	Prorated	
fire safety equipment	Material		f			€200.00		€0.00	Prorated	
workshop materials	Material		w			€750.00		€0.00	Prorated	
wellbeing tracking tools	Material		w			€100.00		€0.00	Prorated	
ergonomic toolsd	Material		e			€400.00		€0.00	Prorated	
first aid station	Material		f			€700.00		€0.00	Prorated	
incident report sw	Material		i			€50.00		€0.00	Prorated	
communication tools	Material		c			€15.00		€0.00	Prorated	
training sw	Material		t			€350.00		€0.00	Prorated	
budgeting tools	Material		b			€76.00		€0.00	Prorated	
video conferencing tools	Material		v			€35.00		€0.00	Prorated	
project management tools	Material		p			€13.00		€0.00	Prorated	
helpdesk sw	Material		h			€40.00		€0.00	Prorated	

Figure 7.2 HR Resource Allocation Table

Taking the feedback from the project proposal interview, this table was from our initial draft. To properly balance the staff distribution, and make effective use of resources, HR staff allocation would need to be reduced from 12 or so down to 4 as it would happen in a business scenario.

Staff Table

Resource Name	Type	Pay/hr	link
Project Manager	Work	€37	
IOT Solutions Architect	Work	€46	
Site Engineer	Work	€38	
Design Engineer	Work	€15	
Electrician	Work	€41	
Plumber	Work	€52	
IT technician	Work	€17	
Compliance Officer	Work	€23	
Systems Engineer	Work	€30	
IOT Specialist	Work	€37	
Software Engineer	Work	€19	
Energy Engineer	Work	€36	
Kitchen Manager	Work	€37	
Structural Engineer	Work	€24	
HVAC Engineer	Work	€30	
Cleaner	Work	€32	
Maintenance Engineer	Work	€15	
Training Manager	Work	€22	

IOT Dishwasher Expert	Work	€45	
Product Specialist	Work	€25	
Support Engineer	Work	€25	
LMS Admin	Work	€33	
UX Designer	Work	€30	
Training Coordinator	Work	€20	
IT admin	Work	€33	
Maintenance Technician	Work	€24	
Maintenance Supervisor	Work	€26	
Logistics manager	Work	€25	
Procurement Specialist	Work	€28	
Transport Coordinator	Work	€18	
Warehouse Manager	Work	€21	
Warehouse Supervisor	Work	€16	
Contracted Driver	Work	€30	
Supply Chain Manager	Work	€22	
Production Manager	Work	€27	
Electrical Engineer	Work	€29	
Assembly Worker	Work	€14	
Accountant	Work	€29	
Procurement Manager	Work	€31	
Operations Manager	Work	€25	
Risk Manager	Work	€40	

Table 7.3 Project Manufacturing and Project Support Staff Allocation Table

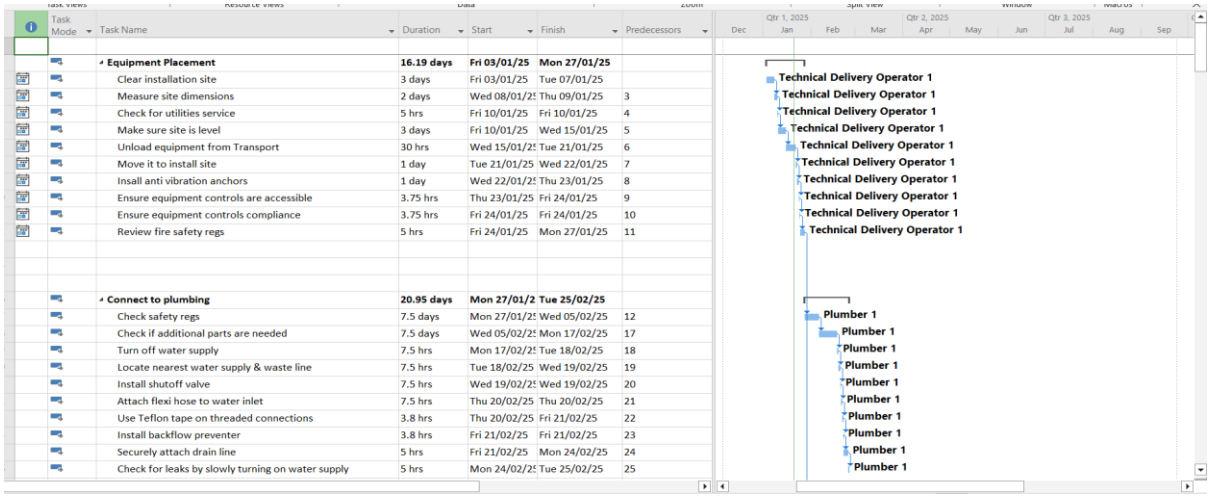
Cost table Year 1 – 5

Resource Name	Hours Worked /yr	Cost / yr	Cost yr 1-5
Project Manager	400	€14,800	€74,000
IOT Solutions Architect	96	€4,416	€22,080
Site Engineer	144	€5,472	€27,360
Design Engineer	240	€3,600	€18,000
Electrician	96	€3,936	€19,680
Plumber	64	€3,328	€16,640
IT technician	64	€1,088	€5,440
Compliance Officer	64	€1,472	€7,360
Systems Engineer	96	€2,880	€14,400
IOT Specialist	224	€4,256	€21,280
Software Engineer	128	€4,608	€23,040
Energy Engineer	64	€2,368	€11,840
Kitchen Manager	32	€768	€3,840
Structural Engineer	32	€960	€4,800
HVAC Engineer	96	€3,072	€15,360
Cleaner	32	€480	€2,400
Maintenance Engineer	32	€704	€3,520
Training Manager	192	€4,984	€24,960

IOT Dishwasher Expert	192	€8,640	€43,200
Product Specialist	208	€5,200	€26,000
Support Engineer	320	€8,000	€40,000
LMS Admin	96	€3,168	€15,840
UX Designer	48	€1,440	€7,200
Training Coordinator	32	€640	€3,200
IT admin	208	€6,864	€34,320
Maintenance Technician	80	€1,920	€9,600
Maintenance Supervisor	32	€832	€4,160
Logistics manager	160	€4,000	€20,000
Procurement Specialist	64	€1,792	€8,960
Transport Coordinator	32	€576	€2,880
Warehouse Manager	160	€3,360	€16,800
Warehouse Supervisor	128	€2,048	€10,240
Contracted Driver	64	€1,920	€9,600
Supply Chain Manager	176	€3,872	€19,360
Production Manager	160	€4,320	€21,600
Electrical Engineer	208	€6,032	€30,160
Assembly Worker	208	€2,912	€14,560
Accountant	144	€4,176	€20,880
Procurement Manager	112	€3,472	€17,360
Operations Manager	128	€3,200	€16,000
Risk Manager	64	€2,560	€12,800
Total Wages Cost Year 1 - 5			€682,480

Table 7.4 Wage Costs for Project Support and Manufacturing for the full duration of the project.

Resource Name	%	Work	Overtime	Baseline	Variance	Actual	Remaining
Technical Delivery Operator 1	0%	215.25 hrs	0 hrs	0 hrs	215.25 hrs	0 hrs	215.25 hrs
Plumber 1	0%	336.35 hrs	0 hrs	0 hrs	336.35 hrs	0 hrs	336.35 hrs
Electrician 1	0%	395 hrs	0 hrs	0 hrs	395 hrs	0 hrs	395 hrs
IT Technician 1	0%	372.5 hrs	0 hrs	0 hrs	372.5 hrs	0 hrs	372.5 hrs
QA Inspector 1	0%	234.64 hrs	0 hrs	0 hrs	234.63 hrs	0 hrs	234.64 hrs
Marketing assistant 1	0%	1,055.5 hrs	0 hrs	0 hrs	1,055.5 hrs	0 hrs	1,055.5 hrs
Marketing manager	0%	825 hrs	0 hrs	0 hrs	825 hrs	0 hrs	825 hrs



Task Mode	Task Name	Duration	Start	Finish	Predecessors
	Equipment Placement	**16.19 days**	**Fri 03/01/25**	**Mon 27/01/25**	
	Clear installation site	3 days	Fri 03/01/25	Tue 07/01/25	
	Measure site dimensions	2 days	Wed 08/01/25	Thu 09/01/25	3
	Check for utilities service	5 hrs	Fri 10/01/25	Fri 10/01/25	4
	Make sure site is level	3 days	Fri 10/01/25	Wed 15/01/25	5
	Unload equipment from Transport	30 hrs	Wed 15/01/25	Tue 21/01/25	6
	Move it to install site	1 day	Tue 21/01/25	Wed 22/01/25	7
	Insall anti vibration anchors	1 day	Wed 22/01/25	Thu 23/01/25	8
	Ensure equipment controls are accessible	3.75 hrs	Thu 23/01/25	Fri 24/01/25	9
	Ensure equipment controls compliance	3.75 hrs	Fri 24/01/25	Fri 24/01/25	10
	Review fire safety regs	5 hrs	Fri 24/01/25	Mon 27/01/25	11
	Connect to plumbing	**20.95 days**	**Mon 27/01/2**	**Tue 25/02/25**	
	Check safety regs	7.5 days	Mon 27/01/25	Wed 05/02/25	12
	Check if additional parts are needed	7.5 days	Wed 05/02/25	Mon 17/02/25	17
	Turn off water supply	7.5 hrs	Mon 17/02/25	Tue 18/02/25	18
	Locate nearest water supply & waste line	7.5 hrs	Tue 18/02/25	Wed 19/02/25	19
	Install shutoff valve	7.5 hrs	Wed 19/02/25	Wed 19/02/25	20
	Attach flexi hose to water inlet	7.5 hrs	Thu 20/02/25	Thu 20/02/25	21
	Use Teflon tape on threaded connections	3.8 hrs	Thu 20/02/25	Fri 21/02/25	22
	Install backflow preventer	3.8 hrs	Fri 21/02/25	Fri 21/02/25	23
	Securely attach drain line	5 hrs	Fri 21/02/25	Mon 24/02/25	24
	Check for leaks by slowly turning on water supply	5 hrs	Mon 24/02/25	Tue 25/02/25	25

Figure 7.5 and figure 7.6 Staff allocation and sample Gantt chart

Resource Name	Type	Std. Rate	hours worked in 2 years	hours worked in 5 years	pay over 5 years	total wage pay
Change Manager	Work	€38.38	96.00	240	9211.2	€351,285.30
Compliance Officer	Work	€29.49	112.00	280	8257.2	
Data Analyst	Work	€25.64	112.00	280	7179.2	
Data Scientist	Work	€34.23	32.00	80	2738.4	
Engineering Team	Work	€43.59	528.00	1320	57538.8	
Facilities Manager	Work	€30.77	96.00	240	7384.8	
Finance Officer	Work	€21.54	32.00	80	1723.2	
Financial Analyst	Work	€30.77	112.00	280	8615.6	
Health & Safety Specialist	Work	€28.21	64.00	160	4513.6	
HR Analyst	Work	€16.77	18.00	45	754.65	
HR assistant	Work	€18.03	48.00	120	2163.6	
HR Communications Specialist	Work	€36.54	2.00	5	182.7	
HR manager	Work	€30.77	64.00	160	4923.2	
HR officer	Work	€20.51	200.00	500	10255	
HR Specialist	Work	€24.36	120.00	300	7308	
HR team	Work	€17.16	96.00	240	4118.4	
HR Trainer	Work	€16.00	576.00	1440	23040	
Innovation Manager	Work	€38.17	16.00	40	1526.8	
Innovation Specialist	Work	€29.66	32.00	80	2372.8	
Instructional Designer	Work	€28.85	32.00	80	2308	
IT Manager	Work	€37.19	80.00	200	7438	
IT Support Team	Work	€28.21	208.00	520	14669.2	
Legal Advisor	Work	€40.53	112.00	280	11348.4	
Line Manager	Work	€20.85	80.00	200	4170	
Manager	Work	€27.80	112.00	280	7784	
Market Research Analyst	Work	€23.72	80.00	200	4744	
Marketing Manager	Work	€28.21	192.00	480	13540.8	
Marketing Specialist	Work	€23.08	80.00	200	4616	
Marketing Team	Work	€18.72	32.00	80	1497.6	
Procurement	Work	€30.58	32.00	80	2446.4	
procurement manager	Work	€37.18	32.00	80	2974.4	
Product Manager	Work	€36.12	208.00	520	18782.4	
Project Manager	Work	€37.03	144.00	360	13330.8	
QA Specialist	Work	€28.21	160.00	400	11284	
Recruitment Manager	Work	€26.92	48.00	120	3230.4	
Risk Management Officer	Work	€30.06	48.00	120	3607.2	
Safety Manager	Work	€33.33	64.00	160	5332.8	
Safety Officer	Work	€25.64	48.00	120	3076.8	
Safety Trainer	Work	€16.92	96.00	240	4060.8	
Team Leads	Work	€30.06	112.00	280	8416.8	
Technical Writer	Work	€28.21	32.00	80	2256.8	
Trainers	Work	€14.56	272.00	680	9900.8	
Training Manager	Work	€27.72	304.00	760	21067.2	
Training Specialist	Work	€28.85	38.00	95	2740.75	
Wellbeing Manager	Work	€30.04	38.00	95	2853.8	

Figure 7.7 HR Resource Allocation

Resource Name	Hours Worked /yr	Cost / yr	Cost yr 1-5
Project Manager	400	€14,800	€74,000
IOT Solutions Architect	96	€4,416	€22,080
Site Engineer	144	€5,472	€27,360
Design Engineer	240	€3,600	€18,000
Electrician	96	€3,936	€19,680

Plumber	64	€3,328	€16,640
IT technician	64	€1,088	€5,440
Compliance Officer	64	€1,472	€7,360
Systems Engineer	96	€2,880	€14,400
IOT Specialist	224	€4,256	€21,280
Software Engineer	128	€4,608	€23,040
Energy Engineer	64	€2,368	€11,840
Kitchen Manager	32	€768	€3,840
Structural Engineer	32	€960	€4,800
HVAC Engineer	96	€3,072	€15,360
Cleaner	32	€480	€2,400
Maintenance Engineer	32	€704	€3,520
Training Manager	192	€4,984	€24,960
IOT Dishwasher Expert	192	€8,640	€43,200
Product Specialist	208	€5,200	€26,000
Support Engineer	320	€8,000	€40,000
LMS Admin	96	€3,168	€15,840
UX Designer	48	€1,440	€7,200
Training Coordinator	32	€640	€3,200
IT admin	208	€6,864	€34,320
Maintenance Technician	80	€1,920	€9,600
Maintenance Supervisor	32	€832	€4,160
Logistics manager	160	€4,000	€20,000
Procurement Specialist	64	€1,792	€8,960
Transport Coordinator	32	€576	€2,880
Warehouse Manager	160	€3,360	€16,800
Warehouse Supervisor	128	€2,048	€10,240
Contracted Driver	64	€1,920	€9,600
Supply Chain Manager	176	€3,872	€19,360
Production Manager	160	€4,320	€21,600
Electrical Engineer	208	€6,032	€30,160
Assembly Worker	208	€2,912	€14,560
Accountant	144	€4,176	€20,880
Procurement Manager	112	€3,472	€17,360
Operations Manager	128	€3,200	€16,000
Risk Manager	64	€2,560	€12,800
Technical Delivery Operator	215	€3,310	€16,552
Plumber	336	€11,116	€55,582
Electrician	395	€11,060	€55,300
IT technician	373	€7,737	€38,684
QA Inspector	235	€3,100	€15,498
Marketing assistant	1,055	€15,041	€75,204

Marketing manager	825	€14,025	€70,125

Resource Name	Hours Worked /yr	Cost / yr	Cost yr 1-5
Project Manager	400	€14,800	€74,000
IOT Solutions Architect	96	€4,416	€22,080
Site Engineer	144	€5,472	€27,360
Design Engineer	240	€3,600	€18,000
Electrician	96	€3,936	€19,680
Plumber	64	€3,328	€16,640
IT technician	64	€1,088	€5,440
Compliance Officer	64	€1,472	€7,360
Systems Engineer	96	€2,880	€14,400
IOT Specialist	224	€4,256	€21,280
Software Engineer	128	€4,608	€23,040
Energy Engineer	64	€2,368	€11,840
Kitchen Manager	32	€768	€3,840
Structural Engineer	32	€960	€4,800
HVAC Engineer	96	€3,072	€15,360
Cleaner	32	€480	€2,400
Maintenance Engineer	32	€704	€3,520
Training Manager	192	€4,984	€24,960
IOT Dishwasher Expert	192	€8,640	€43,200
Product Specialist	208	€5,200	€26,000
Support Engineer	320	€8,000	€40,000
LMS Admin	96	€3,168	€15,840
UX Designer	48	€1,440	€7,200
Training Coordinator	32	€640	€3,200
IT admin	208	€6,864	€34,320
Maintenance Technician	80	€1,920	€9,600
Maintenance Supervisor	32	€832	€4,160
Logistics manager	160	€4,000	€20,000
Procurement Specialist	64	€1,792	€8,960
Transport Coordinator	32	€576	€2,880
Warehouse Manager	160	€3,360	€16,800
Warehouse Supervisor	128	€2,048	€10,240
Contracted Driver	64	€1,920	€9,600
Supply Chain Manager	176	€3,872	€19,360
Production Manager	160	€4,320	€21,600
Electrical Engineer	208	€6,032	€30,160
Assembly Worker	208	€2,912	€14,560
Accountant	144	€4,176	€20,880

Procurement Manager	112	€3,472	€17,360
Operations Manager	128	€3,200	€16,000
Risk Manager	64	€2,560	€12,800
Technical Delivery Operator	215	€3,310	€16,552
Plumber	336	€11,116	€55,582
Electrician	395	€11,060	€55,300
IT technician	373	€7,737	€38,684
QA Inspector	235	€3,100	€15,498
Marketing assistant	1,055	€15,041	€75,204
Marketing manager	825	€14,025	€70,125
Change Manager	48	€1,842.24	€9,211.20
Compliance Officer	56	€1,651.44	€8,257.20
Data Analyst	56	€1,435.84	€7,179.20
Data Scientist	16	€547.68	€2,738.40
Engineering Team	264	€11,507.76	€57,538.80
Facilities Manager	48	€1,476.96	€7,384.80
Finance Officer	16	€344.64	€1,723.20
Financial Analyst	56	€1,723.12	€1,723.20
Health & Safety Specialist	32	€902.72	€8,615.60
HR Analyst	9	€150.93	€4,513.60
HR assistant	24	€432.72	€754.65
HR Communications Specialist	1	€36.54	€182.70
HR manager	32	€984.64	€4,923.20
HR officer	100	€2,051.00	€10,255.00
HR Specialist	60	€1,461.60	€7,308.00
HR team	48	€823.68	€4,118.40
HR Trainer	288	€4,608.00	€23,040.00
Innovation Manager	8	€305.36	€1,526.80
Innovation Specialist	16	€474.56	€2,372.80
Instructional Designer	16	€461.60	€2,308.00
IT Manager	40	€1,487.60	€7,438.00
IT Support Team	104	€2,933.84	€14,669.20
Legal Advisor	56	€2,269.68	€11,348.40
Line Manager	40	€834.00	€4,170.00
Manager	56	€1,556.80	€7,784.00
Market Research Analyst	40	€948.80	€4,744.00
Marketing Manager	96	€2,708.16	€13,540.80
Marketing Specialist	40	€923.20	€4,616.00
Marketing Team	16	€299.52	€1,497.60
Procurement	16	€489.28	€2,446.40
procurement manager	16	€594.88	€2,974.40

Product Manager	104	€3,756.48	€18,782.40
Project Manager	72	€2,666.16	€13,330.80
QA Specialist	80	€2,256.80	€11,284.00
Recruitment Manager	24	€646.08	€3,230.40
Risk Management Officer	24	€721.44	€3,607.20
Safety Manager	32	€1,066.56	€5,332.80
Safety Officer	24	€615.36	€3,076.80
Safety Trainer	48	€812.16	€4,060.80
Team Leads	56	€1,683.36	€8,416.80
Technical Writer	16	€451.36	€2,256.80
Trainers	136	€1,980.16	€9,900.80
Training Manager	152	€4,213.44	€21,067.20
Training Specialist	19	€548.15	€2,740.75
Wellbeing Manager	19	€570.76	€2,853.80

8.0 Cost analysis

Resource Name	Type	Std. R	hours worked in 2 years	hours worked in 5 years	pay over 5 years		total wage pay
Change Manager	Work	€38.38	96.00	240	9211.2		€351,285.30
Compliance Officer	Work	€29.49	112.00	280	8257.2		
Data Analyst	Work	€25.64	112.00	280	7179.2		
Data Scientist	Work	€34.23	32.00	80	2738.4		
Engineering Team	Work	€43.59	528.00	1320	57538.8		
Facilities Manager	Work	€30.77	96.00	240	7384.8		
Finance Officer	Work	€21.54	32.00	80	1723.2		
Financial Analyst	Work	€30.77	112.00	280	8615.6		
Health & Safety Specialist	Work	€28.21	64.00	160	4513.6		
HR Analyst	Work	€16.77	18.00	45	754.65		
HR assistant	Work	€18.03	48.00	120	2163.6		
HR Communications Specialist	Work	€36.54	2.00	5	182.7		
HR manager	Work	€30.77	64.00	160	4923.2		
HR officer	Work	€20.51	200.00	500	10255		
HR Specialist	Work	€24.36	120.00	300	7308		
HR team	Work	€17.16	96.00	240	4118.4		
HR Trainer	Work	€16.00	576.00	1440	23040		
Innovation Manager	Work	€38.17	16.00	40	1526.8		
Innovation Specialist	Work	€29.66	32.00	80	2372.8		
Instructional Designer	Work	€28.85	32.00	80	2308		
IT Manager	Work	€37.19	80.00	200	7438		
IT Support Team	Work	€28.21	208.00	520	14669.2		
Legal Advisor	Work	€40.53	112.00	280	11348.4		
Line Manager	Work	€20.85	80.00	200	4170		
Manager	Work	€27.80	112.00	280	7784		
Market Research Analyst	Work	€23.72	80.00	200	4744		
Marketing Manager	Work	€28.21	192.00	480	13540.8		
Marketing Specialist	Work	€23.08	80.00	200	4616		
Marketing Team	Work	€18.72	32.00	80	1497.6		
Procurement	Work	€30.58	32.00	80	2446.4		
procurement manager	Work	€37.18	32.00	80	2974.4		
Product Manager	Work	€36.12	208.00	520	18782.4		
Project Manager	Work	€37.03	144.00	360	13330.8		
QA Specialist	Work	€28.21	160.00	400	11284		
Recruitment Manager	Work	€26.92	48.00	120	3230.4		
Risk Management Officer	Work	€30.06	48.00	120	3607.2		
Safety Manager	Work	€33.33	64.00	160	5332.8		
Safety Officer	Work	€25.64	48.00	120	3076.8		
Safety Trainer	Work	€16.92	96.00	240	4060.8		
Team Leads	Work	€30.06	112.00	280	8416.8		
Technical Writer	Work	€28.21	32.00	80	2256.8		
Trainers	Work	€14.56	272.00	680	9900.8		
Training Manager	Work	€27.72	304.00	760	21067.2		
Training Specialist	Work	€28.85	38.00	95	2740.75		
Wellbeing Manager	Work	€30.04	38.00	95	2853.8		
				0	0		

Marketing Team	€599.04	€0.00	€599.04	€0.00	€599.04
Change Manager	€3,684.48	€0.00	€3,684.48	€0.00	€3,684.48
Project Manager	€5,332.32	€0.00	€5,332.32	€0.00	€5,332.32
IT Support Team	€2,256.80	€0.00	€2,256.80	€0.00	€2,256.80
Recruitment Manager	€1,292.16	€0.00	€1,292.16	€0.00	€1,292.16
Team Leads	€3,366.72	€0.00	€3,366.72	€0.00	€3,366.72
Marketing Manag	€2,256.80	€0.00	€2,256.80	€0.00	€2,256.80
Market Research Analyst	€4,554.24	€0.00	€4,554.24	€0.00	€4,554.24
Product Manager	€7,512.96	€0.00	€7,512.96	€0.00	€7,512.96
Data Analyst	€2,871.68	€0.00	€2,871.68	€0.00	€2,871.68
Marketing Specia	€738.56	€0.00	€738.56	€0.00	€738.56
Financial Analyst	€3,446.24	€0.00	€3,446.24	€0.00	€3,446.24
Innovation Manager	€1,221.44	€0.00	€1,221.44	€0.00	€1,221.44
Data Scientist	€1,095.36	€0.00	€1,095.36	€0.00	€1,095.36
QA Specialist	€4,513.60	€0.00	€4,513.60	€0.00	€4,513.60
Technical Writer	€902.72	€0.00	€902.72	€0.00	€902.72
Engineering Tear	€23,015.52	€0.00	€23,015.52	€0.00	€23,015.52
Line Manager	€1,668.00	€0.00	€1,668.00	€0.00	€1,668.00
HR team	€274.56	€0.00	€274.56	€0.00	€274.56
Procurement	€0.00	€0.00	€0.00	€0.00	€0.00
Manager	€0.00	€0.00	€0.00	€0.00	€0.00
Innovation Specialist	€2,372.80	€0.00	€2,372.80	€0.00	€2,372.80

8.1 Payback

	Year	Cost / Outgoings	Income
Before the start	0	€813,000	0
	1	€760,590	€1,273,488
	2	€760,590	€1,273,488
	3	€760,590	€1,273,488
	4	€760,590	€1,273,488
	5	€760,590	€1,273,488
Total		€4,615,950	6367440
Pay back	Total cost / Annual Cash flow	€3.62	years

With an initial investment of €813,000 before the project begins. Followed by an average outgoing of €760,590, it leads to a calculated payback period of just over 3 and a half years. With the area of focus for this project, this does not seem unrealistic.

The hospitality / catering industry goes through periods of growth and decline during periods of economic hardship. Overall, though, it is a staple sector around the world. Hence why it was part of our project demographic from a market share perspective.

8.2 Net Present Value

Hurdle Rate	0.1	Companies set as the return required			
inflation	0.02	Real or projected			
Discount rate	0.12	k is Sum of the Hurdle rate plus the inflation			
				Discount Factor	Discounted cash flow
	Year	Cost / Outgoings	Income	1/(1+k)^t	
Before the start	0	-€406,500	0	1	-406,500.00
	1	-€380,295	€636,744	0.892857143	568,521.43
	2	-€380,295	€636,744	0.797193878	507,608.42
	3	-€380,295	€636,744	0.711780248	453,221.80
	4	-€380,295	€636,744	0.635518078	404,662.32
	5	-€380,295	€636,744	0.567426856	361,305.65
Total		-€2,307,975	3183720		
	Present Value of income				2,295,319.62
	Net present value (is the PV income less total outgoings)				1,888,819.62

From running the project initially twice per year to get a general idea of where the project would be at. This when looking at whether it would be worth doing / investing in to allow for the project to happen. Setting what is a realistic hurdle rate and a consistent 2 % rate of inflation per year.

Taking under consideration the wealth of feedback provided to us by the interview review panel where the initial prospect of the project was proposed.

The first piece of feedback was to look at project duration, it would be realistic to run the project 4 times per year. This would double our initial plan of running it twice per year. This in turn would lead to a substantial increase in both overheads and profit margin.

Ending up with the Net Present Value of €1,888,819.62.

These figures would certainly change as the staff reallocation would reduce wage costs in the manufacturing and R&D side of the business when the number of staff in those roles are reduced. However, this wouldn't result in a 1:1 increase in the number of staff in the technical installation side of the business, as we have deemed that there is an excess number of staff in manufacturing and R&D. This decrease in the number of total staff would correlate with a decrease in the total wages, ultimately meaning in an increase in profit margins.

Hurdle Rate	0.1	Companies set as the return required			
inflation	0.02	Real or projected			
Discount rate	0.12	k is Sum of the Hurdle rate plus the inflation			
				Discount Factor	Discounted cash flow
	Year	Cost / Outgoings	Income	1/(1+k)^t	
Before the start	0	-€813,000	0	1	-813,000.00
	1	-€760,590	€1,273,488	0.892857143	1,137,042.86
	2	-€760,590	€1,273,488	0.797193878	1,015,216.84
	3	-€760,590	€1,273,488	0.711780248	906,443.60
	4	-€760,590	€1,273,488	0.635518078	809,324.65
	5	-€760,590	€1,273,488	0.567426856	722,611.29
Total		-€4,615,950	6367440		
	Present Value of income				4,590,639.24
	Net present value (is the PV income less total outgoings)				3,777,639.24

Keeping the hurdle and inflation rate consistent to before. In addition to increasing the needed costs proportionately, it reflects in the initial investment needed, the outgoings, and resulting income.

Comparing the resulting net present value of the new project durations to the prior result. This is a more realistic increase in margin.

8.3 Profitability

The profitability index of the proposed project is calculated by adding our net present value to the initial investment and dividing it by this initial investment.

PI = NPV + Init Inv / nit In.

PI = (€3,777,639.24 + €813,000) / €813,000 = 5.65

Generally, any Profitability over 1 means that the project should go ahead. In the case of this project, the profitability is 5.65 which suggests that it will be successful / profitable over the duration of the project.

9.0 Risk management plans

Accounting for positive and negative risks to the business is important for making sure the business is a long-term success through the 5 year period and beyond. As the hospitality/catering industry is a cyclical industry in terms of revenue growth this is especially important for an IoT enabled commercial kitchen business.

Operational Risks:

These are risks to the business from the day-to-day operations of the business.

Risk: Improper equipment installation can lead to a loss of earnings due to repair costs associated with fixing the improper installation.

> Mitigation: Providing detailed installation instructions to installation technicians, training for the technicians, and conducting thorough post installation quality control checks.

Risk: IoT related vulnerabilities, causing IT failures or security breaches.

> Mitigation: Ensuring strong cyber security practices are followed when conducting initial set up for client, distributing regular OTA (Over the Air) updates to software running on the IoT equipment, performing penetration testing on equipment.

Risk: Insufficient customer acquisition, which would lead to a fall in installations per year and thus a reduction in revenue.

> Mitigation: Conduct accurate market research, develop effective marketing strategies (physical and digital), and adapt equipment offering or services based on customer feedback.

HR and R&D Risks:

These would be risks in the HR and R&D side of the business, and relate to employee specific risks etc.

Risk: Incomplete or inaccurate employee skillset

> Mitigation: Conduct thorough employee interviews, skill reviews, and align recruitment and training processes with job responsibilities.

Risk: Shortage of skilled talent or delayed resource availability.

> Mitigation: Contact and create positive relationships with recruitment agencies and maintain a talent pool for critical roles.

Risk: Subcontractor or external partner performance issues.

> Mitigation: Define clear expectations in contracts, include penalties for delays, and hold regular performance reviews.

Other risks to the business are worker disputes resulting in legal action, sector wide downturn in commercial kitchen equipment demand, and shipping cost increases leading to increased costs of manufacturing and then shipping any parts of the commercial kitchen equipment. These risks can be mitigated by ensuring proper dispute resolution processes are in place within the business, consistent and accurate market sentiment analysis is conducted, and the business maintains a fallback list of suppliers respectively.

10.0 Evaluation methods

To evaluate the business, the need to monitor customer satisfaction, HR, R&D, and technical performance is important. These methods ensure that there is proper measurement of these business metrics.

Important metrics to monitor:

Customer satisfaction: Usability, remote monitoring functionality, and overall customer feedback.

Profitability: Track and measure the progress towards the 5-year profitability deadline.

Customer demand: Monitor increased demand for IoT enabled commercial kitchens and successful installations.

Employee performance: Measure the efficiency of employees and R&D activities.

Technical Efficiency: Measure the cycle times, energy efficiency, IoT connectivity, and customer satisfaction.

Assessment of prototypes, functional components, and employee effectiveness.

Conduct regular evaluations with HR, R&D, and technical installation teams.

Analyse customer feedback, technical data, and financial results.

Evaluate team performance and overall project success.

10.1 Return on Investment

The return on investment formula is as follows:

Net profit from project / total investment x 100.

Net profit: €1,751,490

Total investment/ outgoings: €4,615,950

RoI = €1,751,490/€4,615,950 x 100 = 37.91 % return on investment.

So, looking at the project this way, even though it takes about 3 and a half years to make back the initial investment / continued outgoings / expenses. If scaled up with multiple simultaneous projects per year, there is a larger profit margin available for the future.

10.2 Comparing projects

To ensure an unbiased and fair comparison, several factors need to be considered. These include the Return on Investment (Roi) as a percentage, the average payback period, and the net present value (NPV) of the investment. Additionally, it is important to compare the industries in which each business operates, as well as the sector and the potential for growth of the product within that market.

10.3 Control

1. Milestone Tracking: Use a Gantt chart to schedule and monitor completion dates and goals.

2. Quality Assurance: Regularly test hardware, software, and IoT features to meet industry standards.

3. Budget Monitoring: Track expenses vs. what was set out in the budget.

4. Risk Management: Update mitigation plans monthly.

5. Change Control: Evaluate scope, cost, and timeline impacts before approving changes.

6. Communication: Hold regular meetings and provide clear updates via reports.

7. IoT Metrics: Monitor uptime, data accuracy, and energy efficiency using real-time tools.

8. Customer Feedback: Gather and act on feedback during early-stage testing.

These controls ensure the project stays on track, within scope, and delivers acceptable outcomes.

Conclusion:

To conclude, this was the proposed solution for an IoT enabled kitchen solution. It covered the Project Overview and what was hoped to be achieved. It covered the approach and stakeholder considerations. Thorough project planning was demonstrated.

Clear financial viability was established from the prior draft. Contingency and risk management were accounted for. When compared to other projects, i.e. the prior proposal. There were lessons learned. Staff rebalancing and reallocation was achieved to reflect more realistic scenarios.

New timeframes were established with multiple project instances throughout the fiscal year. There were comparative metrics used as a benchmark for the proposed solutions.

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